The Architecture Test in Data Centres Around the World

Every node is its own server and computes only with its two neighbours. There is no master and no common clock.

Run of 9 November 2026, 06:10 (Berlin). English edition, complete and unabridged.

The discovery

Space dictates matter. The discovery shows this in a closed architecture of couplings: a closed chain of N points in which every point is tied to exactly two neighbours, L = 2I − A. The order of the whole is carried by this structure itself. No central clock, no master and no outside time are needed for it.

The structure fixes its own size. N = 36 is not chosen. The numbers and the architecture force it, and 36 belongs to the double null.

The difference

Work on coupled chains takes N as an input and observes what a chain does for a chosen N. The discovery turns the question around: N is an outcome, and the structure determines it. A structure that forces its own size is not one more observation of something that happens to be there. It is the order from which what is observed arises. This reverses the direction of explanation in the textbooks: first the structure, then what is built on it.

What this report documents

36 nodes, each on its own server in its own data centre, spread over 31 data centres in Europe, North America, Asia and Australia. Six runs: coupling strength K = 2, 4 and 8, winding 1 and 3, ε = 0.496 close to the limit, and the deletion and recovery of node 17. Every step of every node was recomputed in the browser. The winding stays an integer, with a deviation of at most 2.22e-16 at all 401 time points of the 400-step run. All 47 checks are passed.

Result

47 of 47 checks: passed. Computation ran in 6 runs on 31 data centres, 52600 requests to the nodes in total.

Run	Checks	Result
1. Phase run with sine, K = 4, winding 1, 36 nodes, 400 steps	8 of 8	**passed**
2. Phase run with sine, K = 2, winding 1, 36 nodes, 40 steps	8 of 8	**passed**
3. Phase run with sine, K = 8, winding 1, 36 nodes, 40 steps	8 of 8	**passed**
4. Phase run with sine, K = 4, winding 3, 36 nodes, 40 steps	8 of 8	**passed**
5. Run near the limit, ε = 0.496, 36 nodes, 40 steps	8 of 8	**passed**
6. Failure run: failure and recovery of node 17 at step 20, 36 nodes, 40 steps	7 of 7	**passed**

Procedure, identical in every run

1. The page sends the start command to the 36 nodes. After that it controls nothing.
2. Each node determines which data centre it runs on and computes the fingerprint of its maths library.
3. Each node sends its initial value to its two neighbours.
4. As soon as a node has the values of both neighbours for the current step, it computes the next value and passes it on. There is no clock: each node waits only for its neighbours.
5. Each node stores all values it has computed and received. Contradictions between two copies of the same value would be counted.
6. When all nodes have reached the last step, the page collects all histories, recomputes every step in the browser and checks against the predictions fixed in the code beforehand.

Run 1: Phase run with sine, K = 4, winding 1, 36 nodes, 400 steps

Rule per node and step: θ ← θ + dt·(ω + K·(sin(θ_right − θ) + sin(θ_left − θ))) with K = 4, dt = 0.05, ω per node = 0.02·((7·i mod 13) − mean). Start: uniform twist with winding q = 1, θ_i(0) = 2π·q·i/36, no noise. 400 steps.

Checks

Check	Measured	Result
Winding 1 at all 401 time points (deviation < 1e-9)	largest deviation 2.22e-16	**passed**
Every node at step 400 within 1e-11 of the reference	0.00e+0	**passed**
Every node within 1e-11 of the free re-run in the browser	1.11e-16	**passed**
Neighbour differences at step 400 within the band [0.13; 0.21]	0.132557 to 0.206536	**passed**
Mean of the phases changes by less than 1e-12	1.78e-15	**passed**
Every step recomputed from the server values, deviation at most 1e-11	14400 calculations, of which bit-identical 14395, largest difference 1.11e-16	**passed**
400 steps on all 36 nodes	400	**passed**
No node reported a contradiction between received values	0	**passed**

The checksum of the final state is 397e95fd3638f30b2ccadc38a70ba143a11e13980d99948caf47572d1a94f06b; it depends on the maths library (sine).

Initial values, intermediate states and final state of all nodes

Node	Step 0	Step 1	Step 2	Step 100	Step 200	Step 400
0	0.00000000000000	-0.00586111111111112	-0.00817571901419180	-0.0443045548170348	-0.0576905950224496	-0.0681069260774986
1	0.174532925199433	0.175671814088322	0.174250446910622	0.131626291465641	0.120241064308104	0.111047012177880
2	0.349065850398866	0.344204739287755	0.341903884243233	0.300961053455804	0.291985934246513	0.284297291685044
3	0.523598775598299	0.525737664487188	0.525316297309488	0.494391872218785	0.488120763764807	0.482177266889718
4	0.698131700797732	0.694270589686621	0.692969734642099	0.676547387271233	0.673112236328530	0.669082546834372
5	0.872664625997165	0.875803514886054	0.876382147708354	0.878216434192207	0.877618158268587	0.875618519556318
6	1.04719755119660	1.04433644008549	1.04403558504096	1.06396969788270	1.06605028295410	1.06611675007783
7	1.22173047639603	1.22586936528492	1.22744799810722	1.26458682180925	1.26908699758795	1.27120334940084
8	1.39626340159546	1.39440229048435	1.39510143543983	1.44458977858784	1.45114849242375	1.45522243829714
9	1.57079632679490	1.57593521568379	1.57851384850609	1.63464948961315	1.64287228334084	1.64876675746372
10	1.74532925199433	1.74446814088322	1.74616728583870	1.79927396903710	1.80875094309206	1.81626724335493
11	1.91986217719376	1.92600106608265	1.92957969890495	1.96896753956455	1.97932744819718	1.98823376567210
12	2.09439510239320	2.09453399128208	2.09467288017097	2.10829405698778	2.11921874807791	2.12924147562914
13	2.26892802759263	2.26306691648152	2.25976606143699	2.24758962997973	2.25883427235309	2.26967953154296
14	2.44346095279206	2.44459984168095	2.44317847450325	2.41718026290292	2.42856766829799	2.43992492062493
15	2.61799387799149	2.61313276688038	2.61083191183586	2.58160018070433	2.59300294972038	2.60454608970268
16	2.79252680319093	2.79466569207982	2.79424432490212	2.77129951255531	2.78261447827773	2.79401331569975
17	2.96705972839036	2.96319861727925	2.96189776223473	2.95070902273965	2.96182623951755	2.97275430107864
18	3.14159265358979	3.14473154247868	3.14531017530098	3.15038576134966	3.16115679073068	3.17129544095276
19	3.31612557878923	3.31326446767811	3.31296361263359	3.33471301206625	3.34493251434951	3.35398705837671
20	3.49065850398866	3.49479739287755	3.49637602569985	3.53427910750175	3.54367023657279	3.55136608031032
21	3.66519142918809	3.66333031807698	3.66402946303246	3.71344816905141	3.72167790883513	3.72778409493451
22	3.83972435438752	3.84486324327641	3.84744187609871	3.90273475589964	3.90942283078645	3.91374129723019
23	4.01425727958696	4.01339616847585	4.01509531343132	4.06650760471055	4.07127571037953	4.07366867246449
24	4.18879020478639	4.19492909367528	4.19850772649758	4.23512214893182	4.23761297901567	4.23798748464154
25	4.36332312998582	4.36346201887471	4.36360090776360	4.37299687226016	4.37293605022236	4.37126970484508
26	4.53785605518526	4.53199494407415	4.52869408902962	4.51029760692607	4.50750334484549	4.50382693107783
27	4.71238898038469	4.71352786927358	4.71210650209588	4.67715597539631	4.67156220016387	4.66596162369207
28	4.88692190558412	4.88206079447301	4.87975993942849	4.83791982520771	4.82961899753445	4.82225420012068
29	5.06145483078356	5.06359371967244	5.06317235249474	5.02280679958921	5.01202290759883	5.00310539508067
30	5.23598775598299	5.23212664487188	5.23082578982735	5.19606501149280	5.18317350249815	5.17297900646630
31	5.41052068118242	5.41365957007131	5.41423820289361	5.38801742640837	5.37349839803174	5.36234174026303
32	5.58505360638185	5.58219249527074	5.58189164022622	5.56293330167144	5.54736665070881	5.53560402760905
33	5.75958653158129	5.76372542047018	5.76530405329248	5.75123323395652	5.73524997589984	5.72325647886354
34	5.93411945678072	5.93225834566961	5.93295749062509	5.91731441999880	5.90157056831025	5.88973106407754
35	6.10865238198015	6.11379127086904	6.11538365654990	6.09166339707336	6.07680294342460	6.06549692502596

Nodes: data centre, requests, library

Node	Data centre	Requests	Contradictions	Library checksum
0	Chicago, USA	1078	0	`50eb6adc87675964…`
1	Osaka, Japan	947	0	`50eb6adc87675964…`
2	Ashburn near Washington, USA	993	0	`50eb6adc87675964…`
3	Prague, Czechia	1075	0	`50eb6adc87675964…`
4	Dallas, USA	901	0	`50eb6adc87675964…`
5	Sydney, Australia	886	0	`50eb6adc87675964…`
6	Amsterdam, Netherlands	925	0	`50eb6adc87675964…`
7	Frankfurt am Main, Germany	855	0	`50eb6adc87675964…`
8	Amsterdam, Netherlands	1027	0	`50eb6adc87675964…`
9	Chicago, USA	1044	0	`50eb6adc87675964…`
10	Tokyo, Japan	899	0	`50eb6adc87675964…`
11	Chicago, USA	880	0	`50eb6adc87675964…`
12	Milan, Italy	1021	0	`50eb6adc87675964…`
13	San Jose, California, USA	912	0	`50eb6adc87675964…`
14	Brisbane, Australia	857	0	`50eb6adc87675964…`
15	Amsterdam, Netherlands	870	0	`50eb6adc87675964…`
16	Warsaw, Poland	909	0	`50eb6adc87675964…`
17	London, United Kingdom	931	0	`50eb6adc87675964…`
18	Newark, USA	1012	0	`50eb6adc87675964…`
19	Osaka, Japan	954	0	`50eb6adc87675964…`
20	Newark, USA	1042	0	`50eb6adc87675964…`
21	Frankfurt am Main, Germany	983	0	`50eb6adc87675964…`
22	Dallas, USA	980	0	`50eb6adc87675964…`
23	Sydney, Australia	886	0	`50eb6adc87675964…`
24	Amsterdam, Netherlands	953	0	`50eb6adc87675964…`
25	Warsaw, Poland	920	0	`50eb6adc87675964…`
26	Amsterdam, Netherlands	1024	0	`50eb6adc87675964…`
27	Atlanta, USA	1036	0	`50eb6adc87675964…`
28	Osaka, Japan	989	0	`50eb6adc87675964…`
29	Miami, USA	1008	0	`50eb6adc87675964…`
30	Frankfurt am Main, Germany	956	0	`50eb6adc87675964…`
31	San Jose, California, USA	977	0	`50eb6adc87675964…`
32	Melbourne, Australia	908	0	`50eb6adc87675964…`
33	Marseille, France	926	0	`50eb6adc87675964…`
34	Prague, Czechia	959	0	`50eb6adc87675964…`
35	London, United Kingdom	1030	0	`50eb6adc87675964…`

Every step in detail

Per step: winding, largest deviation of the mean of all phases, smallest and largest neighbour difference and the largest deviation between the server values and the recomputation in the browser.

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
0	1.000000000000	0.00e+0	0.174533	0.174533	0.00e+0
1	1.000000000000	0.00e+0	0.163533	0.181533	0.00e+0
2	1.000000000000	0.00e+0	0.159626	0.183412	0.00e+0
3	1.000000000000	0.00e+0	0.157613	0.184206	0.00e+0
4	1.000000000000	4.44e-16	0.156236	0.184769	0.00e+0
5	1.000000000000	4.44e-16	0.155152	0.185283	0.00e+0
6	1.000000000000	0.00e+0	0.154243	0.185786	0.00e+0
7	1.000000000000	0.00e+0	0.153458	0.186287	0.00e+0
8	1.000000000000	0.00e+0	0.152768	0.186784	0.00e+0
9	1.000000000000	4.44e-16	0.152153	0.187278	0.00e+0
10	1.000000000000	0.00e+0	0.151602	0.187765	0.00e+0
11	1.000000000000	4.44e-16	0.151104	0.188244	0.00e+0
12	1.000000000000	4.44e-16	0.150652	0.188712	0.00e+0
13	1.000000000000	0.00e+0	0.150241	0.189170	0.00e+0
14	1.000000000000	0.00e+0	0.149864	0.189615	0.00e+0
15	1.000000000000	0.00e+0	0.149519	0.190046	0.00e+0
16	1.000000000000	4.44e-16	0.149203	0.190465	0.00e+0
17	1.000000000000	0.00e+0	0.148912	0.190869	0.00e+0
18	1.000000000000	0.00e+0	0.148645	0.191258	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
19	1.000000000000	0.00e+0	0.148399	0.191634	0.00e+0
20	1.000000000000	0.00e+0	0.147966	0.191996	0.00e+0
21	1.000000000000	4.44e-16	0.147545	0.192343	0.00e+0
22	1.000000000000	0.00e+0	0.147145	0.192677	0.00e+0
23	1.000000000000	0.00e+0	0.146766	0.192997	0.00e+0
24	1.000000000000	0.00e+0	0.146406	0.193304	0.00e+0
25	1.000000000000	0.00e+0	0.146062	0.193598	0.00e+0
26	1.000000000000	4.44e-16	0.145735	0.193880	0.00e+0
27	1.000000000000	4.44e-16	0.145423	0.194150	0.00e+0
28	1.000000000000	4.44e-16	0.145126	0.194409	0.00e+0
29	1.000000000000	4.44e-16	0.144841	0.194656	0.00e+0
30	1.000000000000	4.44e-16	0.144569	0.194893	0.00e+0
31	1.000000000000	0.00e+0	0.144309	0.195119	0.00e+0
32	1.000000000000	0.00e+0	0.144059	0.195336	0.00e+0
33	1.000000000000	0.00e+0	0.143820	0.195542	0.00e+0
34	1.000000000000	0.00e+0	0.143591	0.195740	0.00e+0
35	1.000000000000	0.00e+0	0.143371	0.195929	0.00e+0
36	1.000000000000	0.00e+0	0.143160	0.196110	0.00e+0
37	1.000000000000	0.00e+0	0.142957	0.196282	0.00e+0
38	1.000000000000	0.00e+0	0.142762	0.196447	0.00e+0
39	1.000000000000	0.00e+0	0.142574	0.196605	0.00e+0
40	1.000000000000	4.44e-16	0.142393	0.196755	0.00e+0
41	1.000000000000	0.00e+0	0.142219	0.196899	0.00e+0
42	1.000000000000	4.44e-16	0.142051	0.197036	0.00e+0
43	1.000000000000	4.44e-16	0.141888	0.197168	0.00e+0
44	1.000000000000	0.00e+0	0.141732	0.197293	0.00e+0
45	1.000000000000	4.44e-16	0.141581	0.197412	0.00e+0
46	1.000000000000	0.00e+0	0.141435	0.197527	0.00e+0
47	1.000000000000	4.44e-16	0.141294	0.197636	0.00e+0
48	1.000000000000	8.88e-16	0.141158	0.197740	0.00e+0
49	1.000000000000	4.44e-16	0.141026	0.197861	0.00e+0
50	1.000000000000	0.00e+0	0.140898	0.197986	0.00e+0
51	1.000000000000	8.88e-16	0.140774	0.198108	0.00e+0
52	1.000000000000	0.00e+0	0.140654	0.198226	0.00e+0
53	1.000000000000	4.44e-16	0.140537	0.198342	0.00e+0
54	1.000000000000	0.00e+0	0.140424	0.198454	0.00e+0
55	1.000000000000	0.00e+0	0.140315	0.198563	0.00e+0
56	1.000000000000	4.44e-16	0.140208	0.198669	0.00e+0
57	1.000000000000	4.44e-16	0.140105	0.198773	0.00e+0
58	1.000000000000	0.00e+0	0.140004	0.198874	0.00e+0
59	1.000000000000	4.44e-16	0.139906	0.198973	0.00e+0
60	1.000000000000	0.00e+0	0.139811	0.199069	0.00e+0
61	1.000000000000	0.00e+0	0.139718	0.199163	0.00e+0
62	1.000000000000	0.00e+0	0.139628	0.199254	0.00e+0
63	1.000000000000	0.00e+0	0.139540	0.199344	0.00e+0
64	1.000000000000	0.00e+0	0.139454	0.199431	0.00e+0
65	1.000000000000	4.44e-16	0.139370	0.199517	0.00e+0
66	1.000000000000	4.44e-16	0.139288	0.199600	0.00e+0
67	1.000000000000	4.44e-16	0.139209	0.199682	0.00e+0
68	1.000000000000	0.00e+0	0.139131	0.199762	0.00e+0
69	1.000000000000	0.00e+0	0.139055	0.199840	0.00e+0
70	1.000000000000	4.44e-16	0.138980	0.199916	0.00e+0
71	1.000000000000	0.00e+0	0.138907	0.199991	0.00e+0
72	1.000000000000	4.44e-16	0.138836	0.200065	0.00e+0
73	1.000000000000	4.44e-16	0.138767	0.200137	0.00e+0
74	1.000000000000	0.00e+0	0.138698	0.200207	0.00e+0
75	1.000000000000	0.00e+0	0.138632	0.200277	0.00e+0
76	1.000000000000	4.44e-16	0.138566	0.200344	0.00e+0
77	1.000000000000	0.00e+0	0.138502	0.200411	0.00e+0
78	1.000000000000	4.44e-16	0.138439	0.200476	0.00e+0
79	1.000000000000	0.00e+0	0.138378	0.200541	0.00e+0
80	1.000000000000	0.00e+0	0.138317	0.200604	0.00e+0
81	1.000000000000	0.00e+0	0.138258	0.200666	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
82	1.000000000000	0.00e+0	0.138200	0.200726	0.00e+0
83	1.000000000000	0.00e+0	0.138142	0.200786	0.00e+0
84	1.000000000000	4.44e-16	0.138086	0.200845	0.00e+0
85	1.000000000000	0.00e+0	0.138031	0.200903	0.00e+0
86	1.000000000000	4.44e-16	0.137977	0.200959	0.00e+0
87	1.000000000000	4.44e-16	0.137923	0.201015	0.00e+0
88	1.000000000000	0.00e+0	0.137871	0.201070	0.00e+0
89	1.000000000000	0.00e+0	0.137819	0.201124	0.00e+0
90	1.000000000000	0.00e+0	0.137768	0.201178	0.00e+0
91	1.000000000000	0.00e+0	0.137718	0.201230	0.00e+0
92	1.000000000000	4.44e-16	0.137669	0.201282	0.00e+0
93	1.000000000000	0.00e+0	0.137621	0.201333	0.00e+0
94	1.000000000000	0.00e+0	0.137573	0.201383	0.00e+0
95	1.000000000000	0.00e+0	0.137526	0.201432	0.00e+0
96	1.000000000000	0.00e+0	0.137480	0.201481	0.00e+0
97	1.000000000000	0.00e+0	0.137434	0.201529	0.00e+0
98	1.000000000000	4.44e-16	0.137389	0.201576	0.00e+0
99	1.000000000000	0.00e+0	0.137345	0.201623	0.00e+0
100	1.000000000000	0.00e+0	0.137301	0.201669	0.00e+0
101	1.000000000000	0.00e+0	0.137258	0.201714	0.00e+0
102	1.000000000000	8.88e-16	0.137215	0.201759	0.00e+0
103	1.000000000000	0.00e+0	0.137173	0.201803	0.00e+0
104	1.000000000000	1.33e-15	0.137131	0.201847	0.00e+0
105	1.000000000000	0.00e+0	0.137090	0.201890	0.00e+0
106	1.000000000000	0.00e+0	0.137050	0.201933	0.00e+0
107	1.000000000000	4.44e-16	0.137010	0.201975	0.00e+0
108	1.000000000000	0.00e+0	0.136970	0.202016	0.00e+0
109	1.000000000000	4.44e-16	0.136931	0.202057	0.00e+0
110	1.000000000000	0.00e+0	0.136893	0.202098	0.00e+0
111	1.000000000000	4.44e-16	0.136855	0.202138	0.00e+0
112	1.000000000000	0.00e+0	0.136817	0.202177	0.00e+0
113	1.000000000000	4.44e-16	0.136780	0.202216	0.00e+0
114	1.000000000000	4.44e-16	0.136743	0.202255	0.00e+0
115	1.000000000000	0.00e+0	0.136707	0.202293	0.00e+0
116	1.000000000000	0.00e+0	0.136671	0.202331	0.00e+0
117	1.000000000000	0.00e+0	0.136635	0.202368	0.00e+0
118	1.000000000000	4.44e-16	0.136600	0.202405	0.00e+0
119	1.000000000000	4.44e-16	0.136565	0.202441	0.00e+0
120	1.000000000000	4.44e-16	0.136531	0.202477	0.00e+0
121	1.000000000000	1.33e-15	0.136497	0.202513	0.00e+0
122	1.000000000000	4.44e-16	0.136463	0.202548	0.00e+0
123	1.000000000000	0.00e+0	0.136430	0.202583	0.00e+0
124	1.000000000000	8.88e-16	0.136397	0.202617	0.00e+0
125	1.000000000000	4.44e-16	0.136364	0.202652	0.00e+0
126	1.000000000000	0.00e+0	0.136332	0.202685	0.00e+0
127	1.000000000000	0.00e+0	0.136300	0.202719	0.00e+0
128	1.000000000000	0.00e+0	0.136268	0.202752	0.00e+0
129	1.000000000000	0.00e+0	0.136237	0.202785	0.00e+0
130	1.000000000000	0.00e+0	0.136206	0.202817	0.00e+0
131	1.000000000000	8.88e-16	0.136175	0.202849	0.00e+0
132	1.000000000000	0.00e+0	0.136144	0.202881	6.94e-18
133	1.000000000000	4.44e-16	0.136114	0.202913	0.00e+0
134	1.000000000000	0.00e+0	0.136084	0.202944	0.00e+0
135	1.000000000000	0.00e+0	0.136055	0.202975	0.00e+0
136	1.000000000000	8.88e-16	0.136025	0.203005	0.00e+0
137	1.000000000000	4.44e-16	0.135996	0.203035	0.00e+0
138	1.000000000000	4.44e-16	0.135967	0.203065	0.00e+0
139	1.000000000000	4.44e-16	0.135939	0.203095	0.00e+0
140	1.000000000000	8.88e-16	0.135910	0.203125	0.00e+0
141	1.000000000000	1.33e-15	0.135882	0.203154	0.00e+0
142	1.000000000000	8.88e-16	0.135855	0.203183	0.00e+0
143	1.000000000000	0.00e+0	0.135827	0.203211	0.00e+0
144	1.000000000000	8.88e-16	0.135800	0.203240	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
145	1.000000000000	1.33e-15	0.135773	0.203268	0.00e+0
146	1.000000000000	0.00e+0	0.135746	0.203296	0.00e+0
147	1.000000000000	8.88e-16	0.135719	0.203323	0.00e+0
148	1.000000000000	8.88e-16	0.135693	0.203351	0.00e+0
149	1.000000000000	4.44e-16	0.135666	0.203378	0.00e+0
150	1.000000000000	0.00e+0	0.135640	0.203405	0.00e+0
151	1.000000000000	4.44e-16	0.135615	0.203431	0.00e+0
152	1.000000000000	0.00e+0	0.135589	0.203458	0.00e+0
153	1.000000000000	1.33e-15	0.135564	0.203484	0.00e+0
154	1.000000000000	4.44e-16	0.135539	0.203510	0.00e+0
155	1.000000000000	4.44e-16	0.135514	0.203536	0.00e+0
156	1.000000000000	4.44e-16	0.135489	0.203561	0.00e+0
157	1.000000000000	8.88e-16	0.135464	0.203587	0.00e+0
158	1.000000000000	0.00e+0	0.135440	0.203612	0.00e+0
159	1.000000000000	4.44e-16	0.135416	0.203637	0.00e+0
160	1.000000000000	0.00e+0	0.135392	0.203661	0.00e+0
161	1.000000000000	8.88e-16	0.135368	0.203686	0.00e+0
162	1.000000000000	4.44e-16	0.135345	0.203710	0.00e+0
163	1.000000000000	0.00e+0	0.135321	0.203734	0.00e+0
164	1.000000000000	4.44e-16	0.135298	0.203758	0.00e+0
165	1.000000000000	0.00e+0	0.135275	0.203782	0.00e+0
166	1.000000000000	8.88e-16	0.135252	0.203805	0.00e+0
167	1.000000000000	4.44e-16	0.135229	0.203829	0.00e+0
168	1.000000000000	8.88e-16	0.135207	0.203852	0.00e+0
169	1.000000000000	1.33e-15	0.135185	0.203875	0.00e+0
170	1.000000000000	8.88e-16	0.135162	0.203897	0.00e+0
171	1.000000000000	4.44e-16	0.135140	0.203920	0.00e+0
172	1.000000000000	4.44e-16	0.135119	0.203942	0.00e+0
173	1.000000000000	0.00e+0	0.135097	0.203965	0.00e+0
174	1.000000000000	4.44e-16	0.135075	0.203987	0.00e+0
175	1.000000000000	4.44e-16	0.135054	0.204009	0.00e+0
176	1.000000000000	0.00e+0	0.135033	0.204030	0.00e+0
177	1.000000000000	4.44e-16	0.135012	0.204052	0.00e+0
178	1.000000000000	0.00e+0	0.134991	0.204073	0.00e+0
179	1.000000000000	4.44e-16	0.134970	0.204095	0.00e+0
180	1.000000000000	1.33e-15	0.134950	0.204116	0.00e+0
181	1.000000000000	4.44e-16	0.134929	0.204137	0.00e+0
182	1.000000000000	4.44e-16	0.134909	0.204157	0.00e+0
183	1.000000000000	8.88e-16	0.134889	0.204178	0.00e+0
184	1.000000000000	8.88e-16	0.134869	0.204198	0.00e+0
185	1.000000000000	4.44e-16	0.134849	0.204219	0.00e+0
186	1.000000000000	4.44e-16	0.134829	0.204239	0.00e+0
187	1.000000000000	4.44e-16	0.134810	0.204259	0.00e+0
188	1.000000000000	0.00e+0	0.134790	0.204279	0.00e+0
189	1.000000000000	0.00e+0	0.134771	0.204298	0.00e+0
190	1.000000000000	8.88e-16	0.134752	0.204318	0.00e+0
191	1.000000000000	8.88e-16	0.134733	0.204337	0.00e+0
192	1.000000000000	0.00e+0	0.134714	0.204357	0.00e+0
193	1.000000000000	4.44e-16	0.134695	0.204376	0.00e+0
194	1.000000000000	4.44e-16	0.134676	0.204395	0.00e+0
195	1.000000000000	0.00e+0	0.134658	0.204414	0.00e+0
196	1.000000000000	0.00e+0	0.134640	0.204432	0.00e+0
197	1.000000000000	4.44e-16	0.134621	0.204451	0.00e+0
198	1.000000000000	0.00e+0	0.134603	0.204469	0.00e+0
199	1.000000000000	0.00e+0	0.134585	0.204488	0.00e+0
200	1.000000000000	4.44e-16	0.134567	0.204506	0.00e+0
201	1.000000000000	4.44e-16	0.134550	0.204524	0.00e+0
202	1.000000000000	0.00e+0	0.134532	0.204542	0.00e+0
203	1.000000000000	4.44e-16	0.134514	0.204560	0.00e+0
204	1.000000000000	4.44e-16	0.134497	0.204577	0.00e+0
205	1.000000000000	0.00e+0	0.134480	0.204595	0.00e+0
206	1.000000000000	8.88e-16	0.134463	0.204612	0.00e+0
207	1.000000000000	1.78e-15	0.134446	0.204630	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
208	1.000000000000	0.00e+0	0.134429	0.204647	0.00e+0
209	1.000000000000	0.00e+0	0.134412	0.204664	0.00e+0
210	1.000000000000	0.00e+0	0.134395	0.204681	0.00e+0
211	1.000000000000	4.44e-16	0.134379	0.204698	0.00e+0
212	1.000000000000	0.00e+0	0.134362	0.204715	0.00e+0
213	1.000000000000	4.44e-16	0.134346	0.204731	0.00e+0
214	1.000000000000	4.44e-16	0.134330	0.204748	0.00e+0
215	1.000000000000	4.44e-16	0.134313	0.204764	0.00e+0
216	1.000000000000	4.44e-16	0.134297	0.204780	0.00e+0
217	1.000000000000	8.88e-16	0.134281	0.204797	0.00e+0
218	1.000000000000	0.00e+0	0.134266	0.204813	0.00e+0
219	1.000000000000	0.00e+0	0.134250	0.204829	0.00e+0
220	1.000000000000	0.00e+0	0.134234	0.204844	1.39e-17
221	1.000000000000	0.00e+0	0.134219	0.204860	0.00e+0
222	1.000000000000	4.44e-16	0.134203	0.204876	0.00e+0
223	1.000000000000	0.00e+0	0.134188	0.204891	0.00e+0
224	1.000000000000	0.00e+0	0.134173	0.204907	0.00e+0
225	1.000000000000	4.44e-16	0.134158	0.204922	0.00e+0
226	1.000000000000	0.00e+0	0.134143	0.204937	0.00e+0
227	1.000000000000	0.00e+0	0.134128	0.204952	0.00e+0
228	1.000000000000	4.44e-16	0.134113	0.204967	0.00e+0
229	1.000000000000	0.00e+0	0.134098	0.204982	0.00e+0
230	1.000000000000	0.00e+0	0.134084	0.204997	0.00e+0
231	1.000000000000	4.44e-16	0.134069	0.205012	0.00e+0
232	1.000000000000	0.00e+0	0.134055	0.205026	0.00e+0
233	1.000000000000	0.00e+0	0.134040	0.205041	0.00e+0
234	1.000000000000	0.00e+0	0.134026	0.205055	0.00e+0
235	1.000000000000	0.00e+0	0.134012	0.205070	0.00e+0
236	1.000000000000	1.33e-15	0.133998	0.205084	0.00e+0
237	1.000000000000	0.00e+0	0.133984	0.205098	0.00e+0
238	1.000000000000	0.00e+0	0.133970	0.205112	0.00e+0
239	1.000000000000	4.44e-16	0.133956	0.205126	0.00e+0
240	1.000000000000	0.00e+0	0.133942	0.205140	0.00e+0
241	1.000000000000	0.00e+0	0.133929	0.205154	0.00e+0
242	1.000000000000	0.00e+0	0.133915	0.205167	0.00e+0
243	1.000000000000	0.00e+0	0.133902	0.205181	0.00e+0
244	1.000000000000	4.44e-16	0.133889	0.205195	0.00e+0
245	1.000000000000	0.00e+0	0.133875	0.205208	0.00e+0
246	1.000000000000	0.00e+0	0.133862	0.205221	0.00e+0
247	1.000000000000	0.00e+0	0.133849	0.205235	0.00e+0
248	1.000000000000	0.00e+0	0.133836	0.205248	0.00e+0
249	1.000000000000	0.00e+0	0.133823	0.205261	0.00e+0
250	1.000000000000	4.44e-16	0.133810	0.205274	0.00e+0
251	1.000000000000	0.00e+0	0.133797	0.205287	0.00e+0
252	1.000000000000	0.00e+0	0.133785	0.205300	0.00e+0
253	1.000000000000	4.44e-16	0.133772	0.205312	0.00e+0
254	1.000000000000	4.44e-16	0.133760	0.205325	0.00e+0
255	1.000000000000	0.00e+0	0.133747	0.205338	0.00e+0
256	1.000000000000	0.00e+0	0.133735	0.205350	0.00e+0
257	1.000000000000	4.44e-16	0.133722	0.205362	0.00e+0
258	1.000000000000	8.88e-16	0.133710	0.205375	0.00e+0
259	1.000000000000	8.88e-16	0.133698	0.205387	0.00e+0
260	1.000000000000	0.00e+0	0.133686	0.205399	0.00e+0
261	1.000000000000	4.44e-16	0.133674	0.205411	0.00e+0
262	1.000000000000	0.00e+0	0.133662	0.205423	0.00e+0
263	1.000000000000	4.44e-16	0.133650	0.205435	0.00e+0
264	1.000000000000	0.00e+0	0.133638	0.205447	0.00e+0
265	1.000000000000	0.00e+0	0.133627	0.205459	1.11e-16
266	1.000000000000	4.44e-16	0.133615	0.205471	0.00e+0
267	1.000000000000	4.44e-16	0.133604	0.205482	0.00e+0
268	1.000000000000	0.00e+0	0.133592	0.205494	0.00e+0
269	1.000000000000	0.00e+0	0.133581	0.205505	0.00e+0
270	1.000000000000	8.88e-16	0.133569	0.205517	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
271	1.000000000000	0.00e+0	0.133558	0.205528	0.00e+0
272	1.000000000000	0.00e+0	0.133547	0.205540	0.00e+0
273	1.000000000000	4.44e-16	0.133536	0.205551	0.00e+0
274	1.000000000000	4.44e-16	0.133525	0.205562	0.00e+0
275	1.000000000000	0.00e+0	0.133514	0.205573	0.00e+0
276	1.000000000000	4.44e-16	0.133503	0.205584	0.00e+0
277	1.000000000000	8.88e-16	0.133492	0.205595	0.00e+0
278	1.000000000000	4.44e-16	0.133481	0.205606	0.00e+0
279	1.000000000000	0.00e+0	0.133470	0.205617	0.00e+0
280	1.000000000000	4.44e-16	0.133460	0.205627	0.00e+0
281	1.000000000000	0.00e+0	0.133449	0.205638	0.00e+0
282	1.000000000000	4.44e-16	0.133439	0.205649	0.00e+0
283	1.000000000000	0.00e+0	0.133428	0.205659	0.00e+0
284	1.000000000000	4.44e-16	0.133418	0.205670	0.00e+0
285	1.000000000000	0.00e+0	0.133408	0.205680	0.00e+0
286	1.000000000000	0.00e+0	0.133397	0.205690	0.00e+0
287	1.000000000000	4.44e-16	0.133387	0.205701	0.00e+0
288	1.000000000000	0.00e+0	0.133377	0.205711	0.00e+0
289	1.000000000000	0.00e+0	0.133367	0.205721	0.00e+0
290	1.000000000000	0.00e+0	0.133357	0.205731	0.00e+0
291	1.000000000000	0.00e+0	0.133347	0.205741	0.00e+0
292	1.000000000000	4.44e-16	0.133337	0.205751	0.00e+0
293	1.000000000000	4.44e-16	0.133327	0.205761	0.00e+0
294	1.000000000000	4.44e-16	0.133317	0.205771	0.00e+0
295	1.000000000000	4.44e-16	0.133308	0.205781	0.00e+0
296	1.000000000000	0.00e+0	0.133298	0.205790	0.00e+0
297	1.000000000000	0.00e+0	0.133288	0.205800	0.00e+0
298	1.000000000000	8.88e-16	0.133279	0.205810	0.00e+0
299	1.000000000000	0.00e+0	0.133269	0.205819	0.00e+0
300	1.000000000000	0.00e+0	0.133260	0.205829	0.00e+0
301	1.000000000000	4.44e-16	0.133251	0.205838	0.00e+0
302	1.000000000000	8.88e-16	0.133241	0.205848	0.00e+0
303	1.000000000000	4.44e-16	0.133232	0.205857	0.00e+0
304	1.000000000000	8.88e-16	0.133223	0.205866	0.00e+0
305	1.000000000000	4.44e-16	0.133214	0.205875	0.00e+0
306	1.000000000000	4.44e-16	0.133205	0.205884	0.00e+0
307	1.000000000000	0.00e+0	0.133196	0.205893	0.00e+0
308	1.000000000000	4.44e-16	0.133187	0.205902	0.00e+0
309	1.000000000000	4.44e-16	0.133178	0.205911	0.00e+0
310	1.000000000000	4.44e-16	0.133169	0.205920	0.00e+0
311	1.000000000000	0.00e+0	0.133160	0.205929	0.00e+0
312	1.000000000000	8.88e-16	0.133151	0.205938	0.00e+0
313	1.000000000000	4.44e-16	0.133143	0.205947	0.00e+0
314	1.000000000000	0.00e+0	0.133134	0.205956	0.00e+0
315	1.000000000000	8.88e-16	0.133126	0.205964	0.00e+0
316	1.000000000000	0.00e+0	0.133117	0.205973	0.00e+0
317	1.000000000000	4.44e-16	0.133109	0.205981	0.00e+0
318	1.000000000000	0.00e+0	0.133100	0.205990	0.00e+0
319	1.000000000000	8.88e-16	0.133092	0.205998	0.00e+0
320	1.000000000000	0.00e+0	0.133083	0.206007	0.00e+0
321	1.000000000000	0.00e+0	0.133075	0.206015	0.00e+0
322	1.000000000000	0.00e+0	0.133067	0.206023	0.00e+0
323	1.000000000000	1.33e-15	0.133059	0.206031	0.00e+0
324	1.000000000000	0.00e+0	0.133051	0.206040	0.00e+0
325	1.000000000000	0.00e+0	0.133043	0.206048	0.00e+0
326	1.000000000000	8.88e-16	0.133034	0.206056	0.00e+0
327	1.000000000000	0.00e+0	0.133027	0.206064	0.00e+0
328	1.000000000000	4.44e-16	0.133019	0.206072	0.00e+0
329	1.000000000000	0.00e+0	0.133011	0.206080	0.00e+0
330	1.000000000000	4.44e-16	0.133003	0.206088	0.00e+0
331	1.000000000000	4.44e-16	0.132995	0.206095	0.00e+0
332	1.000000000000	0.00e+0	0.132987	0.206103	0.00e+0
333	1.000000000000	0.00e+0	0.132980	0.206111	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
334	1.000000000000	0.00e+0	0.132972	0.206119	0.00e+0
335	1.000000000000	0.00e+0	0.132964	0.206126	0.00e+0
336	1.000000000000	0.00e+0	0.132957	0.206134	0.00e+0
337	1.000000000000	4.44e-16	0.132949	0.206142	0.00e+0
338	1.000000000000	0.00e+0	0.132942	0.206149	0.00e+0
339	1.000000000000	0.00e+0	0.132934	0.206157	0.00e+0
340	1.000000000000	0.00e+0	0.132927	0.206164	0.00e+0
341	1.000000000000	8.88e-16	0.132920	0.206171	0.00e+0
342	1.000000000000	4.44e-16	0.132912	0.206179	0.00e+0
343	1.000000000000	4.44e-16	0.132905	0.206186	0.00e+0
344	1.000000000000	4.44e-16	0.132898	0.206193	0.00e+0
345	1.000000000000	4.44e-16	0.132891	0.206200	0.00e+0
346	1.000000000000	4.44e-16	0.132884	0.206208	0.00e+0
347	1.000000000000	0.00e+0	0.132877	0.206215	0.00e+0
348	1.000000000000	0.00e+0	0.132870	0.206222	0.00e+0
349	1.000000000000	0.00e+0	0.132863	0.206229	0.00e+0
350	1.000000000000	8.88e-16	0.132856	0.206236	0.00e+0
351	1.000000000000	4.44e-16	0.132849	0.206243	0.00e+0
352	1.000000000000	4.44e-16	0.132842	0.206250	0.00e+0
353	1.000000000000	4.44e-16	0.132835	0.206256	0.00e+0
354	1.000000000000	0.00e+0	0.132828	0.206263	0.00e+0
355	1.000000000000	4.44e-16	0.132822	0.206270	0.00e+0
356	1.000000000000	4.44e-16	0.132815	0.206277	1.11e-16
357	1.000000000000	0.00e+0	0.132808	0.206283	0.00e+0
358	1.000000000000	4.44e-16	0.132802	0.206290	0.00e+0
359	1.000000000000	4.44e-16	0.132795	0.206297	0.00e+0
360	1.000000000000	4.44e-16	0.132789	0.206303	0.00e+0
361	1.000000000000	4.44e-16	0.132782	0.206310	0.00e+0
362	1.000000000000	8.88e-16	0.132776	0.206316	0.00e+0
363	1.000000000000	1.33e-15	0.132769	0.206323	0.00e+0
364	1.000000000000	0.00e+0	0.132763	0.206329	0.00e+0
365	1.000000000000	4.44e-16	0.132757	0.206335	0.00e+0
366	1.000000000000	4.44e-16	0.132750	0.206342	0.00e+0
367	1.000000000000	4.44e-16	0.132744	0.206348	0.00e+0
368	1.000000000000	4.44e-16	0.132738	0.206354	0.00e+0
369	1.000000000000	8.88e-16	0.132732	0.206361	0.00e+0
370	1.000000000000	0.00e+0	0.132725	0.206367	0.00e+0
371	1.000000000000	0.00e+0	0.132719	0.206373	0.00e+0
372	1.000000000000	8.88e-16	0.132713	0.206379	0.00e+0
373	1.000000000000	0.00e+0	0.132707	0.206385	0.00e+0
374	1.000000000000	4.44e-16	0.132701	0.206391	0.00e+0
375	1.000000000000	0.00e+0	0.132695	0.206397	0.00e+0
376	1.000000000000	8.88e-16	0.132689	0.206403	0.00e+0
377	1.000000000000	4.44e-16	0.132683	0.206409	0.00e+0
378	1.000000000000	4.44e-16	0.132678	0.206415	0.00e+0
379	1.000000000000	4.44e-16	0.132672	0.206421	0.00e+0
380	1.000000000000	8.88e-16	0.132666	0.206427	0.00e+0
381	1.000000000000	4.44e-16	0.132660	0.206432	0.00e+0
382	1.000000000000	8.88e-16	0.132654	0.206438	0.00e+0
383	1.000000000000	0.00e+0	0.132649	0.206444	0.00e+0
384	1.000000000000	4.44e-16	0.132643	0.206450	0.00e+0
385	1.000000000000	4.44e-16	0.132638	0.206455	0.00e+0
386	1.000000000000	0.00e+0	0.132632	0.206461	0.00e+0
387	1.000000000000	8.88e-16	0.132626	0.206466	0.00e+0
388	1.000000000000	8.88e-16	0.132621	0.206472	0.00e+0
389	1.000000000000	4.44e-16	0.132615	0.206477	0.00e+0
390	1.000000000000	0.00e+0	0.132610	0.206483	0.00e+0
391	1.000000000000	8.88e-16	0.132605	0.206488	0.00e+0
392	1.000000000000	4.44e-16	0.132599	0.206494	0.00e+0
393	1.000000000000	8.88e-16	0.132594	0.206499	0.00e+0
394	1.000000000000	1.33e-15	0.132588	0.206505	0.00e+0
395	1.000000000000	0.00e+0	0.132583	0.206510	0.00e+0
396	1.000000000000	0.00e+0	0.132578	0.206515	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
397	1.000000000000	0.00e+0	0.132573	0.206520	0.00e+0
398	1.000000000000	0.00e+0	0.132568	0.206526	0.00e+0
399	1.000000000000	4.44e-16	0.132562	0.206531	0.00e+0
400	1.000000000000	4.44e-16	0.132557	0.206536	0.00e+0

Run 2: Phase run with sine, K = 2, winding 1, 36 nodes, 40 steps

Rule per node and step: θ ← θ + dt·(ω + K·(sin(θ_right − θ) + sin(θ_left − θ))) with K = 2, dt = 0.05, ω per node = 0.02·((7·i mod 13) − mean). Start: uniform twist with winding q = 1, θ_i(0) = 2π·q·i/36, no noise. 40 steps.

Checks

Check	Measured	Result
Winding 1 at all 41 time points (deviation < 1e-9)	largest deviation 2.22e-16	passed
Every node at step 40 within 1e-11 of the reference	0.00e+0	passed
Every node within 1e-11 of the free re-run in the browser	0.00e+0	passed
Neighbour differences at step 40 within the band [0.111; 0.22]	0.121693 to 0.209349	passed
Mean of the phases changes by less than 1e-12	8.88e-16	passed
Every step recomputed from the server values, deviation at most 1e-11	1440 calculations, of which bit-identical 1440, largest difference 0.00e+0	passed
40 steps on all 36 nodes	40	passed
No node reported a contradiction between received values	0	passed

The checksum of the final state is 8142ed78dd012545428aecd59bffab47012bc2ff63cbffcaf44b1287f9bfaa2e; it depends on the maths library (sine).

Initial values, intermediate states and final state of all nodes

Node	Step 0	Step 1	Step 2	Step 10	Step 20	Step 40
0	0.00000000000000	-0.00586111111111112	-0.00994897061820701	-0.0226944220787834	-0.0299874293761588	-0.0406727680910832
1	0.174532925199433	0.175671814088322	0.175530574943916	0.164255676318295	0.152075467210395	0.135943762750143
2	0.349065850398866	0.344204739287755	0.340623756209938	0.325201079970882	0.312310373819685	0.294527210306877
3	0.523598775598299	0.525737664487188	0.526596425342782	0.520744311646508	0.510665010256566	0.495489625162248
4	0.698131700797732	0.694270589686621	0.691689606608804	0.683824824916504	0.677886584331235	0.668638601269596
5	0.872664625997165	0.875803514886054	0.877662275741648	0.879754699484514	0.878750874585529	0.877467818339272
6	1.04719755119660	1.04433644008549	1.04275545700767	1.04288555993031	1.04670173601316	1.05406286631096
7	1.22173047639603	1.22586936528492	1.22872812614051	1.23881940733348	1.24766654291795	1.26330981950302
8	1.39626340159546	1.39440229048435	1.39382130740654	1.40193904065297	1.41525932091763	1.43736958810574
9	1.57079632679490	1.57593521568379	1.57979397653938	1.59775301481339	1.61468373411850	1.63979788318957
10	1.74532925199433	1.74446814088322	1.74488715780540	1.75999045826212	1.77731459473167	1.80018711515545
11	1.91986217719376	1.92600106608265	1.93085982693825	1.95111647571950	1.96346568893775	1.97825009342512
12	2.09439510239320	2.09453399128208	2.09467288017097	2.09578399128208	2.09717288017544	2.09995072230010
13	2.26892802759263	2.26306691648152	2.25848593340370	2.24045150684467	2.23088007147835	2.22165153469687
14	2.44346095279206	2.44459984168095	2.44445860253654	2.43157752430205	2.41703116661771	2.39971548026165
15	2.61799387799149	2.61313276688038	2.60955178380257	2.59381496775078	2.57966203751098	2.56010835858103
16	2.79252680319093	2.79466569207982	2.79552445293541	2.78962894191554	2.77908654531484	2.76254903995612
17	2.96705972839036	2.96319861727925	2.96061763420143	2.95274857558596	2.94668004236979	2.93664621087945
18	3.14159265358979	3.14473154247868	3.14659030333428	3.14868243584450	3.14764943319261	3.14599511649404
19	3.31612557878923	3.31326446767811	3.31168348460030	3.31181357431230	3.31562435474197	3.32283422691154
20	3.49065850398866	3.49479739287755	3.49765615373314	3.50774743457084	3.51659374556478	3.53218313208272
21	3.66519142918809	3.66333031807698	3.66274933499916	3.67086706824125	3.68418724261973	3.70628030015903
22	3.83972435438752	3.84486324327641	3.84872200413201	3.86668104240602	3.88361175042336	3.90872096501162
23	4.01425727958696	4.01339616847585	4.01381518539803	4.02891848585474	4.04624262131179	4.06911375660902
24	4.18879020478639	4.19492909367528	4.19978785453087	4.22004450331213	4.23239371636886	4.24717728425180
25	4.36332312998582	4.36346201887471	4.36360090776360	4.36471201887471	4.36610090653774	4.36887627768604
26	4.53785605518526	4.53199494407415	4.52741396099633	4.50937953443730	4.49980808481443	4.49056966561844
27	4.71238898038469	4.71352786927358	4.71338663012917	4.70050555189034	4.68595905429203	4.66860618057503
28	4.88692190558412	4.88206079447301	4.87847981139519	4.86274299497687	4.84858891391745	4.82890811655025
29	5.06145483078356	5.06359371967245	5.06445248052804	5.05855695535292	5.04800651832271	5.03107244011509
30	5.23598775598299	5.23212664487188	5.22954566179406	5.22167627620195	5.21556062901284	5.20441567450662
31	5.41052068118242	5.41365957007131	5.41551833092690	5.41760537058543	5.41633932525920	5.41188275435136
32	5.58505360638185	5.58219249527074	5.58061151219293	5.58068584905603	5.58354557303246	5.58452185206902
33	5.75958653158129	5.76372542047018	5.76658418132577	5.77622907979832	5.78189788989992	5.78530987457355
34	5.93411945678072	5.93225834566961	5.93167736259179	5.93717448343964	5.94213250432755	5.94383996472476
35	6.10865238198015	6.11379127086904	6.11715690815392	6.12412458183671	6.12419537007232	6.12044233125071

Nodes: data centre, requests, library

Node	Data centre	Requests	Contradictions	Library checksum
0	Atlanta, USA	128	0	`50eb6adc87675964…`
1	Osaka, Japan	120	0	`50eb6adc87675964…`
2	Chicago, USA	120	0	`50eb6adc87675964…`
3	Frankfurt am Main, Germany	125	0	`50eb6adc87675964…`
4	Los Angeles, USA	120	0	`50eb6adc87675964…`
5	Sydney, Australia	118	0	`50eb6adc87675964…`
6	Paris, France	119	0	`50eb6adc87675964…`
7	Warsaw, Poland	118	0	`50eb6adc87675964…`
8	Amsterdam, Netherlands	123	0	`50eb6adc87675964…`
9	Newark, USA	115	0	`50eb6adc87675964…`
10	Singapore	109	0	`50eb6adc87675964…`
11	Atlanta, USA	107	0	`50eb6adc87675964…`
12	Frankfurt am Main, Germany	115	0	`50eb6adc87675964…`
13	Dallas, USA	94	0	`50eb6adc87675964…`
14	Brisbane, Australia	80	0	`50eb6adc87675964…`
15	Amsterdam, Netherlands	84	0	`50eb6adc87675964…`
16	Frankfurt am Main, Germany	85	0	`50eb6adc87675964…`
17	Amsterdam, Netherlands	115	0	`50eb6adc87675964…`
18	Miami, USA	110	0	`50eb6adc87675964…`
19	Osaka, Japan	108	0	`50eb6adc87675964…`
20	Chicago, USA	101	0	`50eb6adc87675964…`
21	Frankfurt am Main, Germany	107	0	`50eb6adc87675964…`
22	San Jose, California, USA	90	0	`50eb6adc87675964…`
23	Melbourne, Australia	87	0	`50eb6adc87675964…`
24	Amsterdam, Netherlands	102	0	`50eb6adc87675964…`
25	Warsaw, Poland	103	0	`50eb6adc87675964…`
26	Amsterdam, Netherlands	136	0	`50eb6adc87675964…`
27	Chicago, USA	133	0	`50eb6adc87675964…`
28	Osaka, Japan	122	0	`50eb6adc87675964…`
29	Miami, USA	122	0	`50eb6adc87675964…`
30	Frankfurt am Main, Germany	118	0	`50eb6adc87675964…`
31	Dallas, USA	109	0	`50eb6adc87675964…`
32	Melbourne, Australia	91	0	`50eb6adc87675964…`
33	Amsterdam, Netherlands	95	0	`50eb6adc87675964…`
34	Frankfurt am Main, Germany	89	0	`50eb6adc87675964…`
35	Amsterdam, Netherlands	109	0	`50eb6adc87675964…`

Every step in detail

Per step: winding, largest deviation of the mean of all phases, smallest and largest neighbour difference and the largest deviation between the server values and the recomputation in the browser.

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
0	1.000000000000	0.00e+0	0.174533	0.174533	0.00e+0
1	1.000000000000	0.00e+0	0.163533	0.181533	0.00e+0
2	1.000000000000	4.44e-16	0.156079	0.185973	0.00e+0
3	1.000000000000	0.00e+0	0.150874	0.188861	0.00e+0
4	1.000000000000	0.00e+0	0.147109	0.190809	0.00e+0
5	1.000000000000	0.00e+0	0.144281	0.192188	0.00e+0
6	1.000000000000	4.44e-16	0.142073	0.193221	0.00e+0
7	1.000000000000	0.00e+0	0.140285	0.194044	0.00e+0
8	1.000000000000	8.88e-16	0.138788	0.194741	0.00e+0
9	1.000000000000	4.44e-16	0.137500	0.195361	0.00e+0
10	1.000000000000	4.44e-16	0.136366	0.195934	0.00e+0
11	1.000000000000	0.00e+0	0.135350	0.196478	0.00e+0
12	1.000000000000	0.00e+0	0.134427	0.197004	0.00e+0
13	1.000000000000	8.88e-16	0.133578	0.197519	0.00e+0
14	1.000000000000	0.00e+0	0.132793	0.198026	0.00e+0
15	1.000000000000	8.88e-16	0.132061	0.198527	0.00e+0
16	1.000000000000	4.44e-16	0.131375	0.199024	0.00e+0
17	1.000000000000	0.00e+0	0.130730	0.199516	0.00e+0
18	1.000000000000	0.00e+0	0.130122	0.200005	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
19	1.000000000000	0.00e+0	0.129547	0.200489	0.00e+0
20	1.000000000000	0.00e+0	0.129003	0.200969	0.00e+0
21	1.000000000000	0.00e+0	0.128485	0.201445	0.00e+0
22	1.000000000000	4.44e-16	0.127993	0.201916	0.00e+0
23	1.000000000000	0.00e+0	0.127525	0.202382	0.00e+0
24	1.000000000000	4.44e-16	0.127079	0.202843	0.00e+0
25	1.000000000000	4.44e-16	0.126653	0.203298	0.00e+0
26	1.000000000000	0.00e+0	0.126246	0.203747	0.00e+0
27	1.000000000000	0.00e+0	0.125858	0.204190	0.00e+0
28	1.000000000000	4.44e-16	0.125486	0.204627	0.00e+0
29	1.000000000000	8.88e-16	0.125130	0.205057	0.00e+0
30	1.000000000000	0.00e+0	0.124790	0.205481	0.00e+0
31	1.000000000000	0.00e+0	0.124464	0.205899	0.00e+0
32	1.000000000000	0.00e+0	0.124151	0.206309	0.00e+0
33	1.000000000000	0.00e+0	0.123851	0.206713	0.00e+0
34	1.000000000000	0.00e+0	0.123564	0.207110	0.00e+0
35	1.000000000000	0.00e+0	0.123289	0.207501	0.00e+0
36	1.000000000000	8.88e-16	0.123024	0.207884	0.00e+0
37	1.000000000000	4.44e-16	0.122771	0.208261	0.00e+0
38	1.000000000000	8.88e-16	0.122528	0.208630	0.00e+0
39	1.000000000000	8.88e-16	0.122134	0.208993	0.00e+0
40	1.000000000000	0.00e+0	0.121693	0.209349	0.00e+0

Run 3: Phase run with sine, K = 8, winding 1, 36 nodes, 40 steps

Rule per node and step: $\theta \leftarrow \theta + dt\cdot(\omega + K\cdot(\sin(\theta_{right} - \theta) + \sin(\theta_{left} - \theta)))$ with K = 8, dt = 0.05, ω per node = 0.02·((7·i mod 13) − mean). Start: uniform twist with winding q = 1, $\theta_i(0) = 2\pi\cdot q\cdot i/36$, no noise. 40 steps.

Checks

Check	Measured	Result
Winding 1 at all 41 time points (deviation < 1e-9)	largest deviation 2.22e-16	**passed**
Every node at step 40 within 1e-11 of the reference	0.00e+0	**passed**
Every node within 1e-11 of the free re-run in the browser	0.00e+0	**passed**
Neighbour differences at step 40 within the band [0.146; 0.198]	0.156384 to 0.187581	**passed**
Mean of the phases changes by less than 1e-12	8.88e-16	**passed**
Every step recomputed from the server values, deviation at most 1e-11	1440 calculations, of which bit-identical 1440, largest difference 0.00e+0	**passed**
40 steps on all 36 nodes	40	**passed**
No node reported a contradiction between received values	0	**passed**

The checksum of the final state is 19662e0d9c68bbbd0f0aa6d369436ca5e2a7b9bdd5888ea3c98a77c24112a942; it depends on the maths library (sine).

Initial values, intermediate states and final state of all nodes

Node	Step 0	Step 1	Step 2	Step 10	Step 20	Step 40
0	0.00000000000000	-0.00586111111111113	-0.00462921580616137	-0.0102439493290658	-0.0142910548068363	-0.0200698587949627
1	0.174532925199433	0.175671814088322	0.171690190844033	0.164686970223266	0.159968202256522	0.154811231108019
2	0.349065850398866	0.344204739287755	0.344464140309822	0.335203483514173	0.330448761105762	0.326313586408298
3	0.523598775598299	0.525737664487188	0.522756041242899	0.516340095757052	0.512481073650655	0.509798878750528
4	0.698131700797732	0.694270589686621	0.695529990708688	0.690619201442146	0.688503607838009	0.687628953720739
5	0.872664625997165	0.875803514886054	0.873821891641764	0.873791619979898	0.874092845569876	0.875210355552059
6	1.04719755119660	1.04433644008549	1.04659584110755	1.04896973211056	1.05186363914104	1.05489886645773
7	1.22173047639603	1.22586936528492	1.22488774204063	1.23226685944851	1.23748337130690	1.24208919271409
8	1.39626340159546	1.39440229048435	1.39766169150642	1.40680598000821	1.41351342531279	1.41909658764592
9	1.57079632679490	1.57593521568379	1.57595359243950	1.58835935895809	1.59541877803663	1.60124564207965
10	1.74532925199433	1.74446814088322	1.74872754190529	1.75935338842190	1.76546190853479	1.77079274238472
11	1.91986217719376	1.92600106608265	1.92701944283836	1.93466145348475	1.93872939843015	1.94296589294759
12	2.09439510239320	2.09453399128208	2.09467288017097	2.09578399128208	2.09717496542272	2.09998705811808
13	2.26892802759263	2.26306691648152	2.26232631750358	2.25690652907941	2.25562280971569	2.25700880672961
14	2.44346095279206	2.44459984168095	2.44061821843666	2.43221459414227	2.42889845178758	2.42918088322829
15	2.61799387799149	2.61313276688038	2.61339216790245	2.60320862360608	2.59895951111779	2.59871610513647
16	2.79252680319093	2.79466569207982	2.79168406883553	2.78476314282802	2.78089903585490	2.78082505345560
17	2.96705972839036	2.96319861727925	2.96445801830132	2.95930772101525	2.95698708003048	2.95773562237653
18	3.14159265358979	3.14473154247868	3.14274991923439	3.14262794235527	3.14269449351211	3.14472850345812
19	3.31612557878923	3.31326446767812	3.31552386870018	3.31786806780152	3.32057928145037	3.32405687910368
20	3.49065850398866	3.49479739287755	3.49381576963326	3.50118828914155	3.50628663893523	3.51102884064767
21	3.66519142918809	3.66333031807698	3.66658971909905	3.67573286732878	3.68237442293340	3.68789099411899
22	3.83972435438752	3.84486324327641	3.84488162003212	3.85728738655072	3.86431294917731	3.86990959236667
23	4.01425727958696	4.01339616847585	4.01765556949791	4.02828141601453	4.03437068534609	4.03928889388032
24	4.18879020478639	4.19492909367528	4.19594747043099	4.20358948107738	4.20763649587182	4.21120277997355
25	4.36332312998582	4.36346201887471	4.36360090776360	4.36471201887471	4.36605821338818	4.36781292727317
26	4.53785605518526	4.53199494407415	4.53125434509621	4.52583455667204	4.52444043864674	4.52419719694250
27	4.71238898038469	4.71352786927358	4.70954624602929	4.70114148054846	4.69756647737954	4.69541220717219
28	4.88692190558412	4.88206079447301	4.88232019549508	4.87212709354613	4.86731802055956	4.86356374494204
29	5.06145483078356	5.06359371967244	5.06061209642816	5.05363957486487	5.04866250389430	5.04373219783382
30	5.23598775598299	5.23212664487188	5.23338604589394	5.22803486203901	5.22368648262227	5.21801403039852
31	5.41052068118242	5.41365957007131	5.41167794682702	5.41092014125598	5.40760680704819	5.40154326011820
32	5.58505360638185	5.58219249527074	5.58445189629281	5.58511196754008	5.58267978335698	5.57645589541942
33	5.75958653158129	5.76372542047018	5.76274379722589	5.76622963543744	5.76420671268818	5.75794532323797
34	5.93411945678072	5.93225834566961	5.93551774669168	5.93674318953466	5.93444217798142	5.92821517857254
35	6.10865238198015	6.11379127086904	6.11183715334187	6.11167410908700	6.10860448054560	6.10250883016445

Nodes: data centre, requests, library

Node	Data centre	Requests	Contradictions	Library checksum
0	Ashburn near Washington, USA	122	0	`50eb6adc87675964…`
1	Tokyo, Japan	119	0	`50eb6adc87675964…`
2	Ashburn near Washington, USA	110	0	`50eb6adc87675964…`
3	Frankfurt am Main, Germany	122	0	`50eb6adc87675964…`
4	Los Angeles, USA	100	0	`50eb6adc87675964…`
5	Sydney, Australia	102	0	`50eb6adc87675964…`
6	Amsterdam, Netherlands	105	0	`50eb6adc87675964…`
7	Prague, Czechia	100	0	`50eb6adc87675964…`
8	London, United Kingdom	101	0	`50eb6adc87675964…`
9	Ashburn near Washington, USA	97	0	`50eb6adc87675964…`
10	Osaka, Japan	70	0	`50eb6adc87675964…`
11	Atlanta, USA	69	0	`50eb6adc87675964…`
12	Frankfurt am Main, Germany	98	0	`50eb6adc87675964…`
13	Dallas, USA	88	0	`50eb6adc87675964…`
14	Melbourne, Australia	71	0	`50eb6adc87675964…`
15	London, United Kingdom	83	0	`50eb6adc87675964…`
16	Milan, Italy	77	0	`50eb6adc87675964…`
17	Paris, France	102	0	`50eb6adc87675964…`
18	Atlanta, USA	111	0	`50eb6adc87675964…`
19	Tokyo, Japan	106	0	`50eb6adc87675964…`
20	Chicago, USA	99	0	`50eb6adc87675964…`
21	Milan, Italy	110	0	`50eb6adc87675964…`
22	Denver, USA	101	0	`50eb6adc87675964…`
23	Sydney, Australia	95	0	`50eb6adc87675964…`
24	Marseille, France	106	0	`50eb6adc87675964…`
25	Milan, Italy	90	0	`50eb6adc87675964…`
26	Amsterdam, Netherlands	92	0	`50eb6adc87675964…`
27	Ashburn near Washington, USA	100	0	`50eb6adc87675964…`
28	Seoul, South Korea	98	0	`50eb6adc87675964…`
29	Newark, USA	111	0	`50eb6adc87675964…`
30	Frankfurt am Main, Germany	108	0	`50eb6adc87675964…`
31	San Jose, California, USA	105	0	`50eb6adc87675964…`
32	Sydney, Australia	97	0	`50eb6adc87675964…`
33	Amsterdam, Netherlands	101	0	`50eb6adc87675964…`
34	Frankfurt am Main, Germany	96	0	`50eb6adc87675964…`
35	Amsterdam, Netherlands	107	0	`50eb6adc87675964…`

Every step in detail

Per step: winding, largest deviation of the mean of all phases, smallest and largest neighbour difference and the largest deviation between the server values and the recomputation in the browser.

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
0	1.000000000000	0.00e+0	0.174533	0.174533	0.00e+0
1	1.000000000000	0.00e+0	0.163533	0.181533	0.00e+0
2	1.000000000000	4.44e-16	0.166719	0.178292	0.00e+0
3	1.000000000000	0.00e+0	0.163287	0.180945	0.00e+0
4	1.000000000000	0.00e+0	0.163996	0.180207	0.00e+0
5	1.000000000000	8.88e-16	0.162629	0.181420	0.00e+0
6	1.000000000000	0.00e+0	0.162626	0.181510	0.00e+0
7	1.000000000000	4.44e-16	0.161980	0.182198	0.00e+0
8	1.000000000000	4.44e-16	0.161807	0.182510	0.00e+0
9	1.000000000000	4.44e-16	0.161449	0.182984	0.00e+0
10	1.000000000000	0.00e+0	0.161123	0.183320	0.00e+0
11	1.000000000000	4.44e-16	0.160716	0.183686	0.00e+0
12	1.000000000000	8.88e-16	0.160354	0.183990	0.00e+0
13	1.000000000000	0.00e+0	0.160023	0.184286	0.00e+0
14	1.000000000000	8.88e-16	0.159724	0.184546	0.00e+0
15	1.000000000000	0.00e+0	0.159450	0.184789	0.00e+0
16	1.000000000000	0.00e+0	0.159200	0.185007	0.00e+0
17	1.000000000000	0.00e+0	0.158970	0.185208	0.00e+0
18	1.000000000000	0.00e+0	0.158758	0.185390	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
19	1.000000000000	4.44e-16	0.158563	0.185556	0.00e+0
20	1.000000000000	0.00e+0	0.158382	0.185707	0.00e+0
21	1.000000000000	4.44e-16	0.158214	0.185845	0.00e+0
22	1.000000000000	0.00e+0	0.158058	0.185970	0.00e+0
23	1.000000000000	8.88e-16	0.157913	0.186084	0.00e+0
24	1.000000000000	0.00e+0	0.157777	0.186188	0.00e+0
25	1.000000000000	0.00e+0	0.157650	0.186297	0.00e+0
26	1.000000000000	4.44e-16	0.157530	0.186415	0.00e+0
27	1.000000000000	0.00e+0	0.157418	0.186526	0.00e+0
28	1.000000000000	8.88e-16	0.157312	0.186632	0.00e+0
29	1.000000000000	0.00e+0	0.157212	0.186733	0.00e+0
30	1.000000000000	0.00e+0	0.157118	0.186828	0.00e+0
31	1.000000000000	4.44e-16	0.157028	0.186919	0.00e+0
32	1.000000000000	0.00e+0	0.156943	0.187006	0.00e+0
33	1.000000000000	4.44e-16	0.156861	0.187089	0.00e+0
34	1.000000000000	0.00e+0	0.156784	0.187168	0.00e+0
35	1.000000000000	4.44e-16	0.156710	0.187244	0.00e+0
36	1.000000000000	0.00e+0	0.156640	0.187317	0.00e+0
37	1.000000000000	0.00e+0	0.156572	0.187387	0.00e+0
38	1.000000000000	4.44e-16	0.156507	0.187454	0.00e+0
39	1.000000000000	4.44e-16	0.156444	0.187519	0.00e+0
40	1.000000000000	0.00e+0	0.156384	0.187581	0.00e+0

Run 4: Phase run with sine, K = 4, winding 3, 36 nodes, 40 steps

Rule per node and step: θ ← θ + dt·(ω + K·(sin(θ_right − θ) + sin(θ_left − θ))) with K = 4, dt = 0.05, ω per node = 0.02·((7·i mod 13) − mean). Start: uniform twist with winding q = 3, θ_i(0) = 2π·q·i/36, no noise. 40 steps.

Checks

Check	Measured	Result
Winding 3 at all 41 time points (deviation < 1e-9)	largest deviation 4.44e-16	passed
Every node at step 40 within 1e-11 of the reference	1.78e-15	passed
Every node within 1e-11 of the free re-run in the browser	0.00e+0	passed
Neighbour differences at step 40 within the band [0.478; 0.559]	0.488304 to 0.548008	passed
Mean of the phases changes by less than 1e-12	1.78e-15	passed
Every step recomputed from the server values, deviation at most 1e-11	1440 calculations, of which bit-identical 1440, largest difference 0.00e+0	passed
40 steps on all 36 nodes	40	passed
No node reported a contradiction between received values	0	passed

The checksum of the final state is b5e199f9512e554e79412ea123950c0b313176681962c1211cae491d894e8b0b; it depends on the maths library (sine).

Initial values, intermediate states and final state of all nodes

Node	Step 0	Step 1	Step 2	Step 10	Step 20	Step 40
0	0.00000000000000	-0.00586111111111114	-0.00860097914355841	-0.0162057884604457	-0.0218031498311719	-0.0303019149224478
1	0.523598775598299	0.524737664487188	0.523625553458537	0.512158440626842	0.503416165560447	0.492821877747887
2	1.04719755119660	1.04233644008549	1.03972632889191	1.02776685459083	1.01807862931096	1.00707990256684
3	1.57079632679490	1.57293521568379	1.57282310465514	1.56468911928331	1.55644755833250	1.54726493105785
4	2.09439510239320	2.09053399128208	2.08892388008851	2.08363218848406	2.07859726179765	2.07332698356796
5	2.61799387799149	2.62113276688038	2.62202065585173	2.62159161093250	2.62080263796381	2.62101050609012
6	3.14159265358979	3.13873154247868	3.13812143128511	3.14077810156309	3.14454082125107	3.15069977555923
7	3.66519142918809	3.66933031807698	3.67121820704833	3.67876797347520	3.68696151208371	3.69853500197398
8	4.18879020478639	4.18692909367528	4.18731898248171	4.19785983226676	4.20961462800983	4.22480035673327
9	4.71238898038469	4.71752786927358	4.72041575824493	4.73531005504619	4.74887474841361	4.76502804084519
10	5.23598775598299	5.23512664487188	5.23651653367831	5.25226123472167	5.26475508129960	5.27879730481019
11	5.75958653158129	5.76572542047018	5.76961330944153	5.78312134816467	5.79118189789741	5.80034712502614
12	6.28318530717959	6.28332419606847	6.28346308495736	6.28457419606848	6.28596308723399	6.28874394081923
13	6.80678408277788	6.80092297166677	6.79731286047320	6.78602704397228	6.78074429056233	6.77714390360431
14	7.33038285837618	7.33152174726507	7.33040963623642	7.31688715741528	7.30717121270297	7.29870543691529
15	7.85398163397448	7.84912052286337	7.84651041166980	7.83383833709076	7.82305212251320	7.81250168354331
16	8.37758040957278	8.37971929846167	8.37960718743302	8.37128856056637	8.36231494236916	8.35278508209339
17	8.90117918517108	8.89731807405997	8.89570796286640	8.89038044513124	8.88497862071882	8.87915272610742
18	9.42477796076938	9.42791684965827	9.42880473862962	9.42837075700050	9.42743512377785	9.42716111076515
19	9.94837673636768	9.94551562525657	9.94490551406300	9.94756171791433	9.95127512891471	9.95710463198396
20	10.4719755119660	10.4761144008549	10.4780022898262	10.4855520297836	10.4937316319726	10.5051128734252
21	10.9955742875643	10.9937131764532	10.9941030652596	11.0046439143485	11.0163953103022	11.0314799868807
22	11.5191730631626	11.5243119520515	11.5271998410228	11.5420941378241	11.5556581298959	11.5717616367494
23	12.0427718387609	12.0419107276498	12.0433006164562	12.0590453174996	12.0715390370127	12.0855526472273
24	12.5663706143592	12.5725095032481	12.5763973922194	12.5899054309426	12.5979659364015	12.6070993796436
25	13.0899693899575	13.0901082788464	13.0902471677352	13.0913582788464	13.0927469830379	13.0954611551677
26	13.6135681655558	13.6077070544447	13.6040969432511	13.5928111267502	13.5875272579333	13.5837648472808
27	14.1371669411541	14.1383058300430	14.1371937190143	14.1236712394962	14.1139494422486	14.1050984963424
28	14.6607657167524	14.6559046056413	14.6532944944477	14.6406223915937	14.6298098569660	14.6184061410861
29	15.1843644923507	15.1865033812396	15.1863912702109	15.1780721048175	15.1689946352521	15.1576919677763
30	15.7079632679490	15.7041021568379	15.7024920456443	15.6971582722146	15.6914038967686	15.6821912896396
31	16.2315620435473	16.2347009324362	16.2355888214075	16.2351040542272	16.2331226604806	16.2268543569800
32	16.7551608191456	16.7522997080345	16.7516895968409	16.7540458765318	16.7551197621728	16.7513031802137
33	17.2787595947439	17.2828984836328	17.2847863726041	17.2909680709284	17.2934464544823	17.2906668741783
34	17.8023583703422	17.8004972592311	17.8008871480375	17.8065764830914	17.8080989903547	17.8045482613379
35	18.3259571459405	18.3310960348294	18.3331136806396	18.3349407121787	18.3333163207639	18.3275271261109

Nodes: data centre, requests, library

Node	Data centre	Requests	Contradictions	Library checksum
0	Atlanta, USA	86	0	`50eb6adc87675964…`
1	Singapore	79	0	`50eb6adc87675964…`
2	Newark, USA	99	0	`50eb6adc87675964…`
3	Frankfurt am Main, Germany	84	0	`50eb6adc87675964…`
4	San Jose, California, USA	81	0	`50eb6adc87675964…`
5	Sydney, Australia	80	0	`50eb6adc87675964…`
6	Madrid, Spain	93	0	`50eb6adc87675964…`
7	Frankfurt am Main, Germany	100	0	`50eb6adc87675964…`
8	Amsterdam, Netherlands	101	0	`50eb6adc87675964…`
9	Newark, USA	100	0	`50eb6adc87675964…`
10	Hong Kong, China (Special Administrative Region)	92	0	`50eb6adc87675964…`
11	Ashburn near Washington, USA	102	0	`50eb6adc87675964…`
12	Vienna, Austria	112	0	`50eb6adc87675964…`
13	San Jose, California, USA	94	0	`50eb6adc87675964…`
14	Sydney, Australia	93	0	`50eb6adc87675964…`
15	Marseille, France	94	0	`50eb6adc87675964…`
16	Stockholm, Sweden	107	0	`50eb6adc87675964…`
17	Amsterdam, Netherlands	105	0	`50eb6adc87675964…`
18	Atlanta, USA	101	0	`50eb6adc87675964…`
19	Tokyo, Japan	88	0	`50eb6adc87675964…`
20	Atlanta, USA	86	0	`50eb6adc87675964…`
21	Warsaw, Poland	98	0	`50eb6adc87675964…`
22	Los Angeles, USA	78	0	`50eb6adc87675964…`
23	Sydney, Australia	85	0	`50eb6adc87675964…`
24	Marseille, France	84	0	`50eb6adc87675964…`
25	Warsaw, Poland	74	0	`50eb6adc87675964…`
26	Amsterdam, Netherlands	88	0	`50eb6adc87675964…`
27	Newark, USA	87	0	`50eb6adc87675964…`
28	Tokyo, Japan	70	0	`50eb6adc87675964…`
29	Chicago, USA	71	0	`50eb6adc87675964…`
30	Vienna, Austria	77	0	`50eb6adc87675964…`
31	Dallas, USA	80	0	`50eb6adc87675964…`
32	Brisbane, Australia	75	0	`50eb6adc87675964…`
33	Lisbon, Portugal	73	0	`50eb6adc87675964…`
34	Bucharest, Romania	91	0	`50eb6adc87675964…`
35	Amsterdam, Netherlands	84	0	`50eb6adc87675964…`

Every step in detail

Per step: winding, largest deviation of the mean of all phases, smallest and largest neighbour difference and the largest deviation between the server values and the recomputation in the browser.

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
0	3.000000000000	0.00e+0	0.523599	0.523599	0.00e+0
1	3.000000000000	1.78e-15	0.512599	0.530599	0.00e+0
2	3.000000000000	0.00e+0	0.507841	0.533097	0.00e+0
3	3.000000000000	1.78e-15	0.505306	0.534213	0.00e+0
4	3.000000000000	0.00e+0	0.503649	0.534904	0.00e+0
5	3.000000000000	0.00e+0	0.502397	0.535465	0.00e+0
6	3.000000000000	0.00e+0	0.501371	0.535985	0.00e+0
7	3.000000000000	0.00e+0	0.500492	0.536493	0.00e+0
8	3.000000000000	1.78e-15	0.499719	0.536996	0.00e+0
9	3.000000000000	0.00e+0	0.499030	0.537495	0.00e+0
10	3.000000000000	1.78e-15	0.498409	0.537990	0.00e+0
11	3.000000000000	1.78e-15	0.497846	0.538480	0.00e+0
12	3.000000000000	1.78e-15	0.497333	0.538962	0.00e+0
13	3.000000000000	0.00e+0	0.496862	0.539437	0.00e+0
14	3.000000000000	1.78e-15	0.496429	0.539902	0.00e+0
15	3.000000000000	0.00e+0	0.496030	0.540356	0.00e+0
16	3.000000000000	1.78e-15	0.495661	0.540800	0.00e+0
17	3.000000000000	1.78e-15	0.495320	0.541232	0.00e+0
18	3.000000000000	0.00e+0	0.495003	0.541653	0.00e+0

Step	Winding	Mean dev.	Neighbour diff. min	Neighbour diff. max	Recomputation
19	3.000000000000	0.00e+0	0.494709	0.542061	0.00e+0
20	3.000000000000	1.78e-15	0.494436	0.542457	0.00e+0
21	3.000000000000	0.00e+0	0.494183	0.542840	0.00e+0
22	3.000000000000	0.00e+0	0.493846	0.543210	0.00e+0
23	3.000000000000	1.78e-15	0.493412	0.543569	0.00e+0
24	3.000000000000	1.78e-15	0.492998	0.543915	0.00e+0
25	3.000000000000	0.00e+0	0.492603	0.544249	0.00e+0
26	3.000000000000	0.00e+0	0.492226	0.544572	0.00e+0
27	3.000000000000	1.78e-15	0.491865	0.544882	0.00e+0
28	3.000000000000	1.78e-15	0.491520	0.545182	0.00e+0
29	3.000000000000	1.78e-15	0.491189	0.545471	0.00e+0
30	3.000000000000	1.78e-15	0.490872	0.545748	0.00e+0
31	3.000000000000	1.78e-15	0.490567	0.546016	0.00e+0
32	3.000000000000	0.00e+0	0.490275	0.546273	0.00e+0
33	3.000000000000	0.00e+0	0.489994	0.546521	0.00e+0
34	3.000000000000	0.00e+0	0.489724	0.546759	0.00e+0
35	3.000000000000	1.78e-15	0.489465	0.546989	0.00e+0
36	3.000000000000	1.78e-15	0.489215	0.547209	0.00e+0
37	3.000000000000	0.00e+0	0.488975	0.547421	0.00e+0
38	3.000000000000	0.00e+0	0.488743	0.547624	0.00e+0
39	3.000000000000	0.00e+0	0.488519	0.547820	0.00e+0
40	3.000000000000	0.00e+0	0.488304	0.548008	0.00e+0

Run 5: Run near the limit, ε = 0.496, 36 nodes, 40 steps

Rule per node and step: $x \leftarrow x + \varepsilon \cdot (\text{right} + \text{left} - 2x)$ with $\varepsilon = 0.496$. Start: $x_i(0) = ((7 \cdot i \bmod 13)/13)$. 40 steps.

Checks

Check	Measured	Result
Every step recomputed from the server values, bit-identical	1440 calculations, of which bit-identical 1440	**passed**
Sum of all values conserved (< 1e-12)	0.00e+0	**passed**
40 steps on all 36 nodes	40	**passed**
No node reported a contradiction between received values	0	**passed**
Decay of the slowest oscillation per step 0.984929 (deviation < 1e-9)	0.984929290988 (target 0.984929290988)	**passed**
Decay of the fastest oscillation per step 0.984 (deviation < 1e-9)	0.984000000000 (target 0.984000000000)	**passed**
Checksum of all steps equal to the prediction	8b0a9a3141029ab6c20c0127c0cf0cd6b3c3bf25de5b86390d0f4afffd3d1d01	**passed**
Checksum of the final state equal to the prediction	a91f85df97fc7259af40911d974e4866fcf61ae50a705ab4b2142e1414b4f097	**passed**

Initial values, intermediate states and final state of all nodes

Node	Step 0	Step 1	Step 2	Step 10	Step 20	Step 40
0	0.00000000000000	0.686769230769231	0.105609846153846	0.185732036339910	0.236775327768031	0.317048405656932
1	0.538461538461538	0.0424615384615384	0.625147076923077	0.642664409045193	0.618494009992246	0.557758123009058
2	0.0769230769230769	0.572923076923077	0.0848590769230769	0.182589165093275	0.246226771931825	0.324845547522903
3	0.615384615384615	0.119384615384615	0.607448615384615	0.640970806563359	0.621661978803486	0.557878977589070
4	0.153846153846154	0.649846153846154	0.161782153846154	0.212960125351095	0.271676783164825	0.341782579907089
5	0.692307692307692	0.196307692307692	0.684371692307692	0.665150552352840	0.621780720353017	0.550736087597664
6	0.230769230769231	0.726769230769231	0.238705230769231	0.270930761940573	0.311714657523558	0.365822415225723
7	0.769230769230769	0.273230769230769	0.761294769230769	0.685302309930927	0.607750096038536	0.535201756234750
8	0.307692307692308	0.803692307692308	0.315628307692308	0.338644444376208	0.360373472699104	0.394484566081143
9	0.846153846153846	0.350153846153846	0.838217846153846	0.650260933273943	0.567667506488308	0.511535135910601
10	0.384615384615385	0.880615384615385	0.392551384615385	0.402945791934704	0.411198538531964	0.425116136270520
11	0.923076923076923	0.427076923076923	0.669124923076923	0.536996340416630	0.501096501312321	0.482339185386149
12	0.461538461538462	0.461538461538462	0.461538461538462	0.461538461538462	0.460976119602569	0.455038842267568
13	0.00000000000000	0.496000000000000	0.253952000000000	0.386080582660293	0.424137534717473	0.452492974479809
14	0.538461538461538	0.0424615384615384	0.530525538461538	0.520131131142219	0.509179344675397	0.481434231137134
15	0.0769230769230769	0.572923076923077	0.0848590769230769	0.273162601661438	0.361499350562306	0.427823266994061
16	0.615384615384615	0.119384615384615	0.607448615384615	0.583933804865644	0.553433986142512	0.501149091922310
17	0.153846153846154	0.649846153846154	0.161782153846154	0.241501952998508	0.331719368391327	0.413127237317818
18	0.692307692307692	0.196307692307692	0.684371692307692	0.646598526124243	0.585270479079336	0.510857279523279
19	0.230769230769231	0.726769230769231	0.238705230769231	0.276478396952680	0.337791901839559	0.410614387702528
20	0.769230769230769	0.273230769230769	0.761294769230769	0.681574970078415	0.591295598516962	0.507863214284882
21	0.307692307692308	0.803692307692308	0.315628307692308	0.339143118211279	0.369565741732454	0.419540529044123
22	0.846153846153846	0.350153846153846	0.838217846153846	0.649914321415485	0.561201858562995	0.491304617151955
23	0.384615384615385	0.880615384615385	0.392551384615385	0.402945791934704	0.413441278181379	0.437018727049835
24	0.923076923076923	0.427076923076923	0.669124923076923	0.536996340416630	0.497315534658131	0.463036730395851
25	0.461538461538462	0.461538461538462	0.461538461538462	0.461538461538462	0.460183220200938	0.459340974437480
26	0.00000000000000	0.496000000000000	0.253952000000000	0.385733970801836	0.416698665606544	0.427517938984050
27	0.538461538461538	0.0424615384615384	0.530525538461538	0.519728614145301	0.505812976408497	0.483070219687018
28	0.0769230769230769	0.572923076923077	0.0848590769230769	0.269289908448927	0.341994515224628	0.390614050779869
29	0.615384615384615	0.119384615384615	0.607448615384615	0.580458981717211	0.547720634339126	0.505573289271563
30	0.153846153846154	0.649846153846154	0.161782153846154	0.221641292773340	0.288082211147304	0.357940293500923
31	0.692307692307692	0.196307692307692	0.684371692307692	0.634502354799323	0.581765064997812	0.525104227822247
32	0.230769230769231	0.726769230769231	0.238705230769231	0.214205497751027	0.256359894567259	0.333598468559525
33	0.769230769230769	0.273230769230769	0.761294769230769	0.663710034420570	0.603914646825704	0.540664314461331
34	0.307692307692308	0.803692307692308	0.221006769230769	0.204390627940643	0.240628705285101	0.319777490538682
35	0.846153846153846	0.159384615384615	0.740544000000000	0.660421809813937	0.614364234896696	0.551717917063789

Nodes: data centre, requests, library

Node	Data centre	Requests	Contradictions	Library checksum
0	Chicago, USA	123	0	`50eb6adc87675964…`
1	Hong Kong, China (Special Administrative Region)	109	0	`50eb6adc87675964…`
2	Ashburn near Washington, USA	106	0	`50eb6adc87675964…`
3	Frankfurt am Main, Germany	107	0	`50eb6adc87675964…`
4	Seattle, USA	97	0	`50eb6adc87675964…`
5	Melbourne, Australia	77	0	`50eb6adc87675964…`
6	London, United Kingdom	94	0	`50eb6adc87675964…`
7	Vienna, Austria	77	0	`50eb6adc87675964…`
8	Amsterdam, Netherlands	84	0	`50eb6adc87675964…`
9	Ashburn near Washington, USA	95	0	`50eb6adc87675964…`
10	Osaka, Japan	101	0	`50eb6adc87675964…`
11	Chicago, USA	108	0	`50eb6adc87675964…`
12	Milan, Italy	116	0	`50eb6adc87675964…`
13	Dallas, USA	93	0	`50eb6adc87675964…`
14	Melbourne, Australia	85	0	`50eb6adc87675964…`
15	Paris, France	96	0	`50eb6adc87675964…`
16	Frankfurt am Main, Germany	84	0	`50eb6adc87675964…`
17	London, United Kingdom	107	0	`50eb6adc87675964…`
18	Atlanta, USA	108	0	`50eb6adc87675964…`
19	Osaka, Japan	104	0	`50eb6adc87675964…`
20	Newark, USA	103	0	`50eb6adc87675964…`
21	Milan, Italy	108	0	`50eb6adc87675964…`
22	Dallas, USA	85	0	`50eb6adc87675964…`
23	Sydney, Australia	83	0	`50eb6adc87675964…`
24	Paris, France	100	0	`50eb6adc87675964…`
25	Prague, Czechia	83	0	`50eb6adc87675964…`
26	Amsterdam, Netherlands	100	0	`50eb6adc87675964…`
27	Ashburn near Washington, USA	107	0	`50eb6adc87675964…`
28	Osaka, Japan	94	0	`50eb6adc87675964…`
29	Ashburn near Washington, USA	93	0	`50eb6adc87675964…`
30	Frankfurt am Main, Germany	117	0	`50eb6adc87675964…`
31	Denver, USA	109	0	`50eb6adc87675964…`
32	Brisbane, Australia	96	0	`50eb6adc87675964…`
33	Amsterdam, Netherlands	106	0	`50eb6adc87675964…`
34	Frankfurt am Main, Germany	104	0	`50eb6adc87675964…`
35	Paris, France	128	0	`50eb6adc87675964…`

Every step in detail

Per step: largest deviation of the sum, smallest and largest value and the largest deviation between the server values and the recomputation in the browser.

Step	Sum dev.	Value min	Value max	Recomputation
0	0.00e+0	0.000000000	0.9230769231	0.00e+0
1	0.00e+0	0.04246153846	0.8806153846	0.00e+0
2	1.11e-16	0.08485907692	0.8382178462	0.00e+0
3	0.00e+0	0.09947376246	0.7958832837	0.00e+0
4	0.00e+0	0.1173234020	0.7700098943	0.00e+0
5	0.00e+0	0.1317944557	0.7583023087	0.00e+0
6	0.00e+0	0.1434194486	0.7312806925	0.00e+0
7	0.00e+0	0.1546786863	0.7218778001	0.00e+0
8	1.11e-16	0.1646624184	0.7073805226	0.00e+0
9	1.11e-16	0.1727166121	0.6908968132	0.00e+0
10	0.00e+0	0.1825891651	0.6853023099	0.00e+0
11	0.00e+0	0.1878286312	0.6719920658	0.00e+0
12	1.11e-16	0.1969425052	0.6656843864	0.00e+0
13	1.11e-16	0.2010671914	0.6585088888	0.00e+0
14	1.11e-16	0.2074193003	0.6488377078	0.00e+0
15	1.11e-16	0.2130540461	0.6457431623	0.00e+0
16	1.11e-16	0.2174738501	0.6398317478	0.00e+0
17	0.00e+0	0.2238785411	0.6337945314	0.00e+0
18	1.11e-16	0.2272433645	0.6307455924	0.00e+0
19	1.11e-16	0.2327622524	0.6263080634	0.00e+0

Step	Sum dev.	Value min	Value max	Recomputation
20	0.00e+0	0.2367753278	0.6217807204	0.00e+0
21	0.00e+0	0.2417073143	0.6189209930	0.00e+0
22	1.11e-16	0.2460743918	0.6151548838	0.00e+0
23	0.00e+0	0.2506099943	0.6114414725	0.00e+0
24	0.00e+0	0.2551273390	0.6084281063	0.00e+0
25	0.00e+0	0.2593869710	0.6044371487	0.00e+0
26	1.11e-16	0.2639158042	0.6016277100	0.00e+0
27	0.00e+0	0.2679744217	0.5979573648	0.00e+0
28	0.00e+0	0.2724223569	0.5948595229	0.00e+0
29	0.00e+0	0.2763254417	0.5915031294	0.00e+0
30	0.00e+0	0.2806330746	0.5881993060	0.00e+0
31	1.11e-16	0.2844070836	0.5851403559	0.00e+0
32	1.11e-16	0.2885383232	0.5817003674	0.00e+0
33	0.00e+0	0.2921975687	0.5789170003	0.00e+0
34	5.55e-17	0.2961326688	0.5753992513	0.00e+0
35	1.11e-16	0.2996838725	0.5728671731	0.00e+0
36	1.11e-16	0.3034144118	0.5693199921	0.00e+0
37	0.00e+0	0.3068597239	0.5670143460	0.00e+0
38	0.00e+0	0.3103849947	0.5634773040	0.00e+0
39	0.00e+0	0.3137239892	0.5613738425	0.00e+0
40	0.00e+0	0.3170484057	0.5578789776	0.00e+0

Run 6: Failure run: failure and recovery of node 17 at step 20, 36 nodes, 40 steps

Rule per node and step: $x \leftarrow x + \varepsilon \cdot (\text{right} + \text{left} - 2x)$ with $\varepsilon = 0.25$. Start: $x_i(0) = ((7 \cdot i \bmod 13)/13)$. 40 steps. At step 20 node 17 is deleted (state and memory empty) and fetches its entire history back from its two neighbours.

Checks

Check	Measured	Result
Every step recomputed from the server values, bit-identical	1440 calculations, of which bit-identical 1440	**passed**
Sum of all values conserved (< 1e-12)	0.00e+0	**passed**
40 steps on all 36 nodes	40	**passed**
No node reported a contradiction between received values	0	**passed**
Checksum of all steps equal to the prediction	37824610b0f8c7ff86a5e9b5da8853c8c689f93ae102eea44e40132b0b3e5e25	**passed**
Checksum of the final state equal to the prediction	48799547bb68ca0f7bf83c79510676397bae820100acfe4d4f04605e2390e008	**passed**
Node 17 was deleted at step 20 and restored its state from both neighbours	restored: 20 values, both copies equal: yes	**passed**

The failure in detail

- At step 20 node 17 deletes its entire state and memory.
- It asks its neighbours, node 16 (left) and node 18 (right), for the values they received from it.
- It receives 20 values from each side (step 0 to 19). Both copies are bit for bit equal.
- It also takes over the histories of its two neighbours as received values and continues computing.
- All other nodes keep running without interruption and wait until node 17 delivers values again.
- The checksum of all steps of all 36 nodes afterwards equals the prediction for the run without failure.

Initial values, intermediate states and final state of all nodes

Node	Step 0	Step 1	Step 2	Step 10	Step 20	Step 40
0	0.00000000000000	0.346153846153846	0.370192307692308	0.408423643845778	0.418253900211382	0.426770957745114
1	0.538461538461538	0.288461538461538	0.312500000000000	0.385277821467473	0.410680198828861	0.428734124987658
2	0.0769230769230769	0.326923076923077	0.326923076923077	0.377533839299129	0.409273956030400	0.432615901535036
3	0.615384615384615	0.365384615384615	0.365384615384615	0.387560257544884	0.416115232240302	0.438679347204476
4	0.153846153846154	0.403846153846154	0.403846153846154	0.411786592923678	0.431065862405866	0.446658406749421
5	0.692307692307692	0.442307692307692	0.442307692307692	0.444398219768818	0.451899423356875	0.455730163955809
6	0.230769230769231	0.480769230769231	0.480769230769231	0.480158512408917	0.474855735316501	0.464645817976813
7	0.769230769230769	0.519230769230769	0.519230769230769	0.514505459712102	0.495388706835981	0.471992057410339
8	0.307692307692308	0.557692307692308	0.557692307692308	0.541807981637808	0.509030470271402	0.476519996959296
9	0.846153846153846	0.596153846153846	0.596153846153846	0.554387092590332	0.512385127136720	0.477459500444666
10	0.384615384615385	0.634615384615385	0.634615384615385	0.544794522798978	0.504146638018868	0.474737084561167
11	0.923076923076923	0.673076923076923	0.610576923076923	0.511175449077900	0.485796691300073	0.469036517995319
12	0.461538461538462	0.461538461538462	0.461538461538462	0.461538461538462	0.461543016919817	0.461678904678586
13	0.00000000000000	0.250000000000000	0.312500000000000	0.411901473999023	0.437298701068147	0.454344725545236
14	0.538461538461538	0.288461538461538	0.288461538461538	0.378282400277945	0.418987630253818	0.448701825516718
15	0.0769230769230769	0.326923076923077	0.326923076923077	0.368690197284405	0.410852766523931	0.446028653078193
16	0.615384615384615	0.365384615384615	0.365384615384615	0.381275690518893	0.414437873879461	0.446923727510958
17	0.153846153846154	0.403846153846154	0.403846153846154	0.408629784217248	0.428509583400647	0.451170301354947
18	0.692307692307692	0.442307692307692	0.442307692307692	0.443231802720290	0.449694994941708	0.457786623438007
19	0.230769230769231	0.480769230769231	0.480769230769231	0.479845120356633	0.473381924119447	0.465248116823244
20	0.769230769230769	0.519230769230769	0.519230769230769	0.514447138859676	0.494567304237115	0.471828935036446
21	0.307692307692308	0.557692307692308	0.557692307692308	0.541801232558030	0.508638787053454	0.475984467760883
22	0.846153846153846	0.596153846153846	0.596153846153846	0.554386725792518	0.512222583691045	0.476687591292568
23	0.384615384615385	0.634615384615385	0.634615384615385	0.544794522798978	0.504081428513368	0.473641217782212
24	0.923076923076923	0.673076923076923	0.610576923076923	0.511175449077900	0.485744798692394	0.467317097313543
25	0.461538461538462	0.461538461538462	0.461538461538462	0.461538461538462	0.461411353997042	0.458811864755238
26	0.00000000000000	0.250000000000000	0.312500000000000	0.411901107201209	0.436888064864643	0.449557754701494
27	0.538461538461538	0.288461538461538	0.288461538461538	0.378274697523851	0.417825087396383	0.440965159314874
28	0.0769230769230769	0.326923076923077	0.326923076923077	0.368612802945651	0.407928801786021	0.434094487167816
29	0.615384615384615	0.365384615384615	0.365384615384615	0.380781100346492	0.407882850461577	0.429447113007076
30	0.153846153846154	0.403846153846154	0.403846153846154	0.406376178448017	0.415361011553054	0.426929835420774
31	0.692307692307692	0.442307692307692	0.442307692307692	0.435454441950871	0.426032219322224	0.425994350203175
32	0.230769230769231	0.480769230769231	0.480769230769231	0.458774786729079	0.435120426007975	0.425899794531267
33	0.769230769230769	0.519230769230769	0.519230769230769	0.468639080341046	0.438985952270951	0.426010045880481
34	0.307692307692308	0.557692307692308	0.533653846153846	0.460876978360690	0.436261546303145	0.426029873249591
35	0.846153846153846	0.500000000000000	0.475961538461538	0.437730202308068	0.428218581558632	0.426106887880791

Nodes: data centre, requests, library

Node	Data centre	Requests	Contradictions	Library checksum
0	Chicago, USA	113	0	`50eb6adc87675964…`
1	Singapore	103	0	`50eb6adc87675964…`
2	Chicago, USA	98	0	`50eb6adc87675964…`
3	Frankfurt am Main, Germany	104	0	`50eb6adc87675964…`
4	Los Angeles, USA	85	0	`50eb6adc87675964…`
5	Sydney, Australia	84	0	`50eb6adc87675964…`
6	Lisbon, Portugal	87	0	`50eb6adc87675964…`
7	Frankfurt am Main, Germany	104	0	`50eb6adc87675964…`
8	Amsterdam, Netherlands	102	0	`50eb6adc87675964…`
9	Atlanta, USA	107	0	`50eb6adc87675964…`
10	Tokyo, Japan	106	0	`50eb6adc87675964…`
11	Miami, USA	96	0	`50eb6adc87675964…`
12	Milan, Italy	104	0	`50eb6adc87675964…`
13	Denver, USA	93	0	`50eb6adc87675964…`
14	Sydney, Australia	97	0	`50eb6adc87675964…`
15	Amsterdam, Netherlands	103	0	`50eb6adc87675964…`
16	Frankfurt am Main, Germany	106	0	`50eb6adc87675964…`
17	Amsterdam, Netherlands	146	0	`50eb6adc87675964…`
18	Miami, USA	113	0	`50eb6adc87675964…`
19	Singapore	114	0	`50eb6adc87675964…`
20	Ashburn near Washington, USA	121	0	`50eb6adc87675964…`
21	Frankfurt am Main, Germany	123	0	`50eb6adc87675964…`
22	San Jose, California, USA	115	0	`50eb6adc87675964…`
23	Melbourne, Australia	105	0	`50eb6adc87675964…`
24	Marseille, France	111	0	`50eb6adc87675964…`
25	Stockholm, Sweden	116	0	`50eb6adc87675964…`
26	Amsterdam, Netherlands	107	0	`50eb6adc87675964…`
27	Miami, USA	102	0	`50eb6adc87675964…`
28	Tokyo, Japan	101	0	`50eb6adc87675964…`
29	Ashburn near Washington, USA	107	0	`50eb6adc87675964…`
30	Milan, Italy	104	0	`50eb6adc87675964…`
31	Seattle, USA	95	0	`50eb6adc87675964…`
32	Sydney, Australia	94	0	`50eb6adc87675964…`
33	London, United Kingdom	103	0	`50eb6adc87675964…`
34	Stockholm, Sweden	95	0	`50eb6adc87675964…`
35	Amsterdam, Netherlands	108	0	`50eb6adc87675964…`

Every step in detail

Per step: largest deviation of the sum, smallest and largest value and the largest deviation between the server values and the recomputation in the browser.

Step	Sum dev.	Value min	Value max	Recomputation
0	0.00e+0	0.000000000	0.9230769231	0.00e+0
1	1.11e-16	0.2500000000	0.6730769231	0.00e+0
2	0.00e+0	0.2884615385	0.6346153846	0.00e+0
3	0.00e+0	0.3040865385	0.6189903846	0.00e+0
4	0.00e+0	0.3197115385	0.6033653846	0.00e+0
5	0.00e+0	0.3333834135	0.5896935096	0.00e+0
6	1.11e-16	0.3432804988	0.5797964243	0.00e+0
7	1.11e-16	0.3499943660	0.5730825571	0.00e+0
8	1.11e-16	0.3565345177	0.5665365366	0.00e+0
9	1.11e-16	0.3627609840	0.5602880625	0.00e+0
10	1.11e-16	0.3686128029	0.5543870926	0.00e+0
11	2.22e-16	0.3740703509	0.5488441724	0.00e+0
12	1.11e-16	0.3791360809	0.5436508426	0.00e+0
13	1.11e-16	0.3838242132	0.5387900288	0.00e+0
14	1.11e-16	0.3881549936	0.5342413190	0.00e+0
15	0.00e+0	0.3921514012	0.5299836055	0.00e+0
16	0.00e+0	0.3958372313	0.5259963621	0.00e+0
17	0.00e+0	0.3992359796	0.5222602146	0.00e+0
18	1.11e-16	0.4023702068	0.5187571507	0.00e+0
19	1.11e-16	0.4052611977	0.5154705498	0.00e+0

Step	Sum dev.	Value min	Value max	Recomputation
20	0.00e+0	0.4078828505	0.5123851271	0.00e+0
21	1.11e-16	0.4097638786	0.5094868406	0.00e+0
22	0.00e+0	0.4115196039	0.5067627871	0.00e+0
23	1.11e-16	0.4131586959	0.5042010986	0.00e+0
24	0.00e+0	0.4146894488	0.5017908447	0.00e+0
25	1.11e-16	0.4161197045	0.4995219445	0.00e+0
26	5.55e-17	0.4174568162	0.4973850857	0.00e+0
27	0.00e+0	0.4187076387	0.4953716540	0.00e+0
28	0.00e+0	0.4198785361	0.4934736695	0.00e+0
29	0.00e+0	0.4209753999	0.4916837300	0.00e+0
30	5.55e-17	0.4220036733	0.4899949610	0.00e+0
31	1.11e-16	0.4227229565	0.4884009712	0.00e+0
32	1.11e-16	0.4232567997	0.4868958120	0.00e+0
33	0.00e+0	0.4237721457	0.4854739422	0.00e+0
34	1.11e-16	0.4242698642	0.4841301956	0.00e+0
35	0.00e+0	0.4247507865	0.4828597522	0.00e+0
36	0.00e+0	0.4252157049	0.4816581116	0.00e+0
37	2.22e-16	0.4254140283	0.4805210699	0.00e+0
38	0.00e+0	0.4255978495	0.4794446977	0.00e+0
39	2.22e-16	0.4257917181	0.4784253204	0.00e+0
40	0.00e+0	0.4258997945	0.4774595004	0.00e+0

Data centres

The 31 data centres on which nodes ran:

- Amsterdam, Netherlands
- Ashburn near Washington, USA
- Atlanta, USA
- Brisbane, Australia
- Bucharest, Romania
- Chicago, USA
- Dallas, USA
- Denver, USA
- Frankfurt am Main, Germany
- Hong Kong, China (Special Administrative Region)
- Lisbon, Portugal
- London, United Kingdom
- Los Angeles, USA
- Madrid, Spain
- Marseille, France
- Melbourne, Australia
- Miami, USA
- Milan, Italy
- Newark, USA
- Osaka, Japan
- Paris, France
- Prague, Czechia
- San Jose, California, USA
- Seattle, USA
- Seoul, South Korea
- Singapore
- Stockholm, Sweden
- Sydney, Australia
- Tokyo, Japan
- Vienna, Austria
- Warsaw, Poland

Maths library of the nodes

Each node computes 14,000 values of sine, cosine, exponential, logarithm, power, arctangent and tangent and reports the SHA-256 checksum of the results. Equal checksums mean the same maths library. Found: 1 distinct.

Checksum (SHA-256)	Nodes
`50eb6adc876759647a24e3b90cd4b903971954abca4e8348f81929479133a7a1`	216

Checksums

Content	SHA-256
Raw data of all runs (file in the repository: weltlauf/ergebnis/ringtest_rohdaten.json)	`ad46baad7560a18ce9be6613c769cbd2dd212342d4dcf31e43a38c613cb704f6`
Short form of this report	`3899f89ca069d39154ace04a1c0c5dca670edd46627cd886ee37a570e954c981`

This English report is rendered from the raw data file of the Cloudflare run of 9 November 2026 with the same analysis code as the German original; the numbers are the same. The German original was generated by the live Cloudflare page.